is the steady-state solution to (9.23), and that the control law for this system corresponding to $\mathcal{J}$ with $N \rightarrow \infty$ is

$$\mathbf{u}(k) = -\mathbf{K}_{\infty}\mathbf{x}(k), \tag{9.47}$$

where, from Eqs. (9.46) and (9.28),

$$\mathbf{K}_{\infty} = (\mathbf{Q}_2 + \mathbf{\Gamma}^T\mathbf{S}_{\infty}\mathbf{\Gamma})^{-1}\mathbf{\Gamma}^T\mathbf{S}_{\infty}\mathbf{\Phi}. \tag{9.48}$$

Furthermore, from Eq. (9.31), the cost associated with using this control law is

$$\mathcal{J}_{\infty} = \frac{1}{2}\mathbf{x}^T(0)\mathbf{S}_{\infty}\mathbf{x}(0).$$

In summary, the complete computational procedure is:

1. Compute eigenvalues of the system matrix $\mathcal{H}_c$ defined by Eq. (9.34).
2. Compute eigenvectors associated with the stable ($|z| < 1$) eigenvalues of $\mathcal{H}_c$ and call them
$$\begin{bmatrix} \mathbf{X}_I \\ \mathbf{\Lambda}_I \end{bmatrix}.$$
3. Compute control gain $\mathbf{K}_{\infty}$ from Eq. (9.48) with $\mathbf{S}_{\infty}$ given by Eq. (9.46).

We have already seen that the stable eigenvalues from Step 1 above are the resulting system closed-loop roots with constant gain $\mathbf{K}_{\infty}$ from Step 3. We can also show that the matrix $\mathbf{X}_I$ of Eq. (9.41) is the matrix of eigenvectors of the optimal steady-state closed-loop system.

Most software packages (see MATLAB's dlqr.m for the discrete case as developed here or lqr.m for the continuous case) use algorithms for these calculations that are closely related to the procedure above. In some cases the software gives the user a choice of the particular method to be used in the solution. Although it is possible to find the LQR gain $\mathbf{K}_{\infty}$ for a SISO system by picking optimal roots from a symmetric root locus and then using Ackermann's formula, it is easier to use the general lqr routines in MATLAB for either SISO or MIMO systems. If a locus of the optimal roots is desired, dlqr.m can be used repetitively for varying values of elements in $\mathbf{Q}_1$ or $\mathbf{Q}_2$.

◆ **Example 9.5** *Optimal Root Locus for the Double Mass-Spring System*

Examine the optimal locus for the double mass–spring system of Appendix A.4 and comment on the relative merits of this approach vs. the pole-placement approach used in Example 8.3 for this system.

Solution. Using the same model and sample period as in Example 8.3, we find values for $\mathbf{\Phi}$, $\mathbf{\Gamma}$ which are then used with dlqr.m to solve for the closed-loop roots for various values of the weighting matrix $\mathbf{Q}_1$. Because the output d is the only quantity of interest, it makes sense to use weighting on that state element only, that is

$$\mathbf{Q}_1 = \begin{bmatrix} q_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

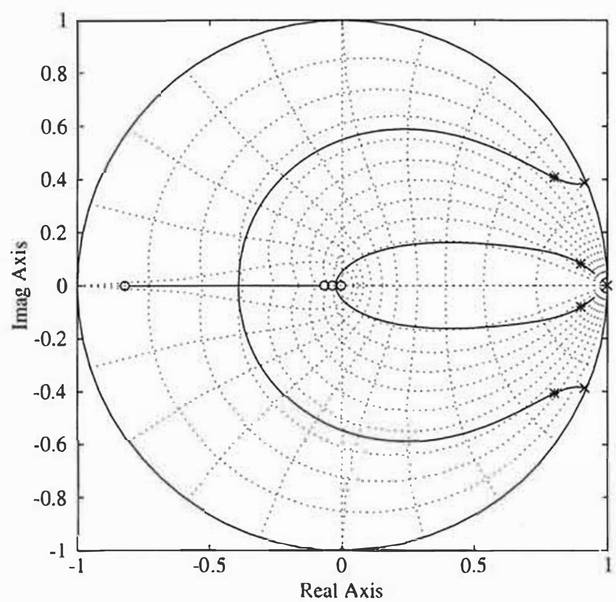
and $\mathbf{Q}_2$ is arbitrarily selected to be 1. The locus is determined by varying the value of q_{11}. The MATLAB script

```
q = logspace(-3,9,300),   Q2=1
for i=1:100,
   Q1=diag([q(i);0;0;0]);
   [k,s,e]=dlqr(phi,gam,Q1,Q2);
end
```

produced a series of values of the closed-loop system roots (e above) which were plotted on the z-plane in Fig. 9.5. Note that the roots selected in Example 8.3 that gave the best results lie at the points marked by stars on the stable portion of the locus. So we see that the optimal solution provides guidance on where to pick the oscillatory poles. In fact, the results from this design led to the selection of the pole locations for the second case in Example 8.3 that yielded the superior response.

◆

Figure 9.5
Locus of optimal roots of Example 9.5



9.3.4 Cost Equivalents

It is sometimes useful to be able to find the discrete cost function defined by Eq. (9.11), which is the equivalent to an analog cost function of the form

$$\mathcal{J}_c = \frac{1}{2}\int_0^{NT} (\mathbf{x}^T\mathbf{Q}_{c1}\mathbf{x} + \mathbf{u}^T\mathbf{Q}_{c2}\mathbf{u})d\tau. \tag{9.49}$$

Having this equivalence relationship will allow us to compute the $\mathcal{J}_c$ of two discrete designs implemented with different sample periods and, therefore, will provide a fair basis for comparison. It will also provide a method for finding a discrete implementation of an optimal, continuous design—an "optimal" version of the emulation design method discussed in Section 7.2. In this section, we will develop the cost equivalence, and the use of it for an emulation design will be discussed in Section 9.3.5.

The integration of the cost in Eq. (9.49) can be broken into sample periods according to

$$\mathcal{J}_c = \sum_{k=0}^{N-1}\frac{1}{2}\int_{kT}^{(k+1)T} (\mathbf{x}^T\mathbf{Q}_{c1}\mathbf{x} + \mathbf{u}^T\mathbf{Q}_{c2}\mathbf{u})d\tau, \tag{9.50}$$

and because

$$\mathbf{x}(kT+\tau) = \mathbf{\Phi}(\tau)\mathbf{x}(kT) + \mathbf{\Gamma}(\tau)\mathbf{u}(kT), \tag{9.51}$$

where

$$\mathbf{\Phi}(\tau) = e^{\mathbf{F}\tau} \qquad \mathbf{\Gamma}(\tau) = \int_0^{\tau} e^{\mathbf{F}\eta}d\eta\mathbf{G}$$

as indicated by Eq. (4.58), substitution of Eq. (9.51) into (9.50) yields

$$\mathcal{J}_c = \frac{1}{2}\sum_{k=0}^{N-1}[(\mathbf{x}^T(k)\mathbf{u}^T(k)]\begin{bmatrix}\mathbf{Q}_{11} & \mathbf{Q}_{12}\\ \mathbf{Q}_{21} & \mathbf{Q}_{22}\end{bmatrix}\begin{bmatrix}\mathbf{x}(k)\\ \mathbf{u}(k)\end{bmatrix}, \tag{9.52}$$

where

$$\begin{bmatrix}\mathbf{Q}_{11} & \mathbf{Q}_{12}\\ \mathbf{Q}_{21} & \mathbf{Q}_{22}\end{bmatrix} = \int_0^T \begin{bmatrix}\mathbf{\Phi}^T(\tau) & \mathbf{0}\\ \mathbf{\Gamma}^T(\tau) & \mathbf{I}\end{bmatrix}\begin{bmatrix}\mathbf{Q}_{c1} & \mathbf{0}\\ \mathbf{0} & \mathbf{Q}_{c2}\end{bmatrix}\begin{bmatrix}\mathbf{\Phi}(\tau) & \mathbf{\Gamma}(\tau)\\ \mathbf{0} & \mathbf{I}\end{bmatrix}d\tau. \tag{9.53}$$

Equation (9.53) is a relationship for the desired equivalent, discrete weighting matrices; however, we see that a complication has arisen in that there are now cross terms that weight the product of $\mathbf{x}$ and $\mathbf{u}$. This can be circumvented by transforming the control to include a linear combination of the state, as we will show below; however, it is also possible to formulate the LQR solution so that it can account for the cross terms.

A method for computing the equivalent gains in Eq. (9.53), due to Van Loan (1978), is to form the matrix exponential[6]

$$\exp\begin{bmatrix} -\mathbf{F}^{\mathrm{T}} & \mathbf{0} & \mathbf{Q}_{c1} & \mathbf{0} \\ -\mathbf{G}^{\mathrm{T}} & \mathbf{0} & \mathbf{0} & \mathbf{Q}_{c2} \\ \mathbf{0} & \mathbf{0} & \mathbf{F} & \mathbf{G} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} T \;\widehat{=}\; \begin{bmatrix} \mathbf{\Phi}_{11} & \mathbf{\Phi}_{12} \\ \mathbf{0} & \mathbf{\Phi}_{22} \end{bmatrix}. \tag{9.54}$$

It will turn out that

$$\mathbf{\Phi}_{22} = \begin{bmatrix} \mathbf{\Phi} & \mathbf{\Gamma} \\ \mathbf{0} & \mathbf{I} \end{bmatrix}$$

$$\mathbf{\Phi}_{11}^{-1} = \mathbf{\Phi}_{22}^{T};$$

however, one needs to calculate the matrix exponential Eq. (9.54) in order to find $\mathbf{\Phi}_{12}$. Because

$$\mathbf{\Phi}_{12} = \mathbf{\Phi}_{11} \begin{bmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{bmatrix},$$

we have the desired result

$$\begin{bmatrix} \mathbf{Q}_{11} & \mathbf{Q}_{12} \\ \mathbf{Q}_{21} & \mathbf{Q}_{22} \end{bmatrix} = \mathbf{\Phi}_{22}^{T}\mathbf{\Phi}_{12}. \tag{9.55}$$

Routines to calculate the discrete equivalent of a continuous cost are available in some of the CAD control design packages (see jdequiv.m in the Digital Control Toolbox). Furthermore, MATLAB has an LQR routine (called lqrd.m) that finds the discrete controller for a continuous cost and computes the necessary discrete cost in the process.

In summary, the continuous cost function $\mathcal{J}_c$ in Eq. (9.49) can be computed from discrete samples of the state and control by transforming the continuous weighting matrices, $\mathbf{Q}_c$'s, according to Eq. (9.55). The resulting discrete weighting matrices include cross terms that weight the product of $\mathbf{x}$ and $\mathbf{u}$. The ability to compute the continuous cost from discrete samples of the state and control is useful for comparing digital controllers of a system with different sample rates and will also be useful in the emulation design method in the next section.

9.3.5 Emulation by Equivalent Cost

The emulation design method discussed in Section 7.2 took the approach that the design of the compensation be done in the continuous domain and the resulting $D(s)$ then be approximated using the digital filtering ideas of Chapter 6. This same approach can be applied when using optimal design methods [see Parsons (1982)]. First, an optimal design iteration is carried out in the continuous domain until the

6 Note that superscript $(\ )^{\mathrm{T}}$ denotes transpose, whereas the entire matrix is multiplied by the sample period, T.

desired specifications are met. The discrete approximation is then obtained by calculating the discrete equivalent of the continuous cost function via Eq. (9.55), and then using that cost in the discrete LQR computation of Table 9.1, with modifications to account for the cross terms in the weighting matrices. These steps are all accomplished by MATLAB's lqrd.m.

◆ **Example 9.6** *Design by Equivalent Cost*

Examine the accuracy of the equivalent cost emulation method for the satellite attitude control example.

Solution. The continuous representation of the system is specified by **F** and **G** from Eq. (4.47). Use of a continuous LQR calculation (lqr.m in MATLAB) with

$$\mathbf{Q}_{c1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$$

and

$$Q_{c2} = [10000,\ 1,\ 0.5,\ 0.2,\ 0.1,\ 0.05,\ 0.02,\ 0.01]$$

results in s-plane roots exactly at $\zeta = 0.7$ as shown by the line in Fig. 9.6. Use of the lqrd.m function computes the discrete controller that minimizes the same continuous cost, thus arriving at an emulation of the continuous design. The equivalent s-plane roots of these digital controllers are also plotted in Fig. 9.6. The figure shows that very little change in root locations occur by this emulation method. In fact the change in root locations is about 1% when

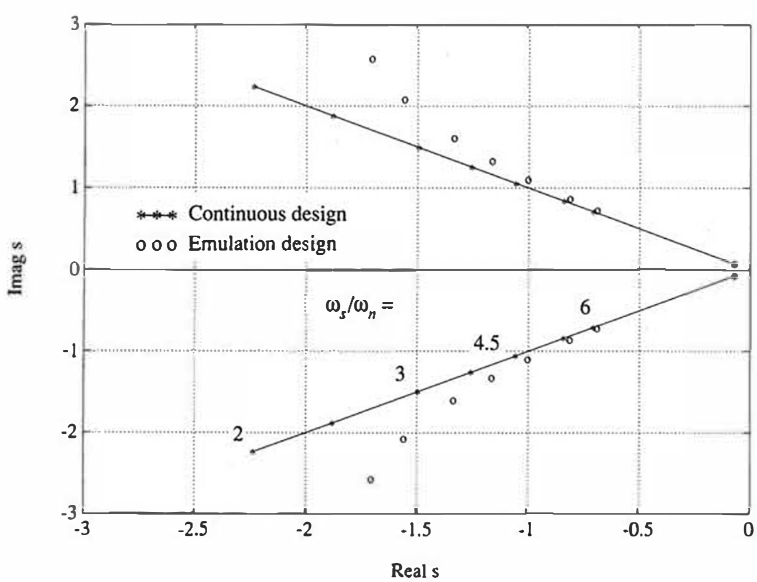


Figure 9.6
Degradations of s-plane root location for the optimal emulation method, Example 9.6

sampling at six times the closed-loop natural frequency ($\omega_s/\omega_n = 6$) and increases to about 10% for $\omega_s/\omega_n = 3$. The accuracy of the zero-pole mapping emulation method was evaluated in Example 7.3; its calculations for a similar example showed that $\omega_s/\omega_n = 30$ resulted in a 10% change in root location, and $\omega_s/\omega_n = 6$ resulted in a 60% reduction in damping.

◆

In general, use of the optimal emulation method will result in a digital controller whose performance will match the continuous design much closer than any of the emulation design methods discussed in Section 7.2 and Chapter 6. A requirement to use the method, however, is that the original continuous design be done using optimal methods so that the continuous weighting matrices $\mathbf{Q}_{c1}$ and $\mathbf{Q}_{c2}$ are available for the conversion because they are the parameters that define the design.

emulation advantages

As discussed in Chapter 7, emulation design is attractive because it allows for the design process to be carried out before specifying the sample rate. Sampling degrades the performance of any system to varying degrees, and it is satisfying to be able to answer how good the control system can be in terms of disturbance rejection, steady-state errors, and so on, before the sampling degradation is introduced. Once the characteristics of a reasonable continuous design are known, the designer is better equipped to select a sample rate with full knowledge of how that selection will affect performance.

We acknowledge that the scenario just presented is not the reality of the typical digital design process. Usually, due to the pressure of schedules, the computer and sample rate are specified long before the controls engineers have a firm grasp on the controller algorithm. Given that reality, the most expedient path is to perform the design directly in the discrete domain and obtain the best possible with that constraint. Furthermore, many design exercises are relatively minor modifications to previous designs. In these cases, too, the most expedient path is to work directly in the discrete domain.

But we maintain that the most desirable design scenario is to gain knowledge of the effects of sampling by first performing the design in the continuous domain, then performing discrete designs. In this case, the emulation method described here is a useful tool to obtain quickly a controller to be implemented digitally or to use as a basis for further refinement in the discrete domain.

9.4 Optimal Estimation

Optimal estimation methods are attractive because they handle multi-output systems easily and allow a designer quickly to determine many good candidate designs of the estimator gain matrix, $\mathbf{L}$. We will first develop the least squares estimation solution for the static case as it is the basis for optimal estimation, then we will extend that to the time-varying optimal estimation solution (commonly known as the "Kalman filter"), and finally show the correspondence between the Kalman filter and the time-varying optimal control solution. Following the same route that we did for the optimal control solution, we will then develop the optimal estimation solution with a steady-state **L**-matrix. In the end, this amounts to another method of computing the **L**-matrix in the equations for the current estimator, Eqs. (8.33) and (8.34), which are

$$\hat{\mathbf{x}}(k) = \bar{\mathbf{x}}(k) + \mathbf{L}(y(k) - \bar{y}(k)), \tag{9.56}$$

where

$$\bar{\mathbf{x}}(k+1) = \mathbf{\Phi}\hat{\mathbf{x}}(k) + \mathbf{\Gamma}\mathbf{u}(k);$$

however, now **L** will be based on minimizing estimation errors rather than picking dynamic characteristics of the estimator error equation.

9.4.1 Least-Squares Estimation

Suppose we have a linear static process given by

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{v}, \tag{9.57}$$

where **y** is a $p \times 1$ measurement vector, **x** is an $n \times 1$ unknown vector, **v** is a $p \times 1$ measurement error vector, and **H** is the matrix relating the measurements to the unknowns. We want to determine the best estimate of **x** given the measurements **y**. Often, the system is overdetermined; that is, there are more measurements in **y** than the unknown vector, **x**. A good way to find the best estimate of **x** is to minimize the sum of the squares of **v**, the fit error. This is called the **least squares** solution. This is both sensible and very convenient analytically. Proceeding, the sum of squares can be written as

$$\mathcal{J} = \frac{1}{2}\mathbf{v}^T\mathbf{v} = \frac{1}{2}(\mathbf{y} - \mathbf{H}\mathbf{x})^T(\mathbf{y} - \mathbf{H}\mathbf{x}) \tag{9.58}$$

and, in order to minimize this expression, we take the derivative with respect to the unknown, that is

$$\frac{\partial \mathcal{J}}{\partial \mathbf{x}} = (\mathbf{y} - \mathbf{H}\mathbf{x})^T(-\mathbf{H}) = 0 \tag{9.59}$$

which results in

$$\mathbf{H}^T\mathbf{y} = \mathbf{H}^T\mathbf{H}\mathbf{x},$$

so that

least squares estimate

$$\hat{\mathbf{x}} = [\mathbf{H}^T\mathbf{H}]^{-1}\mathbf{H}^T\mathbf{y} \tag{9.60}$$

where $\hat{\mathbf{x}}$ designates the "best estimate" of **x**. Note that the matrix to be inverted is $n \times n$ and that p must be $\geq n$ for it to be full rank and the inverse to exist. If there are fewer measurements than unknowns ($p < n$), there are too few measurements to determine a unique value of $\mathbf{x}$.

The difference between the estimate and the actual value of $\mathbf{x}$ is

$$\begin{aligned}\hat{\mathbf{x}} - \mathbf{x} &= [\mathbf{H}^T\mathbf{H}]^{-1}\mathbf{H}^T(\mathbf{H}\mathbf{x} + \mathbf{v}) - \mathbf{x} \\ &= [\mathbf{H}^T\mathbf{H}]^{-1}\mathbf{H}^T\mathbf{v}. \end{aligned} \tag{9.61}$$

estimate accuracy

Equation (9.61) shows that, if $\mathbf{v}$ has zero mean, the error in the estimate, $\hat{\mathbf{x}} - \mathbf{x}$, will also be zero mean, sometimes referred to as an **unbiased estimate**.

The **covariance** of the estimate error, $\mathbf{P}$, is defined to be

$$\begin{aligned}\mathbf{P} &= \mathcal{E}\{(\hat{\mathbf{x}} - \mathbf{x})(\hat{\mathbf{x}} - \mathbf{x})^T\} \\ &= \mathcal{E}\{(\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathbf{v}\mathbf{v}^T\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}\} \\ &= (\mathbf{H}^T\mathbf{H})^{-1}\mathbf{H}^T\mathcal{E}\{\mathbf{v}\mathbf{v}^T\}\mathbf{H}(\mathbf{H}^T\mathbf{H})^{-1}. \end{aligned} \tag{9.62}$$

If the elements in the noise vector, $\mathbf{v}$, are uncorrelated with one another, $\mathcal{E}\{\mathbf{v}\mathbf{v}^T\}$[7] is a diagonal matrix, which we shall call $\mathbf{R}$. Furthermore, if all the elements of $\mathbf{v}$ have the same uncertainty, then all the diagonal elements of $\mathbf{R}$ are identical, and

$$\mathcal{E}\{\mathbf{v}\mathbf{v}^T\} = \mathbf{R} = \mathbf{I}\sigma^2, \tag{9.63}$$

where σ is the rms value of each element in $\mathbf{v}$. In this case, Eq. (9.62) can be written as

$$\mathbf{P} = (\mathbf{H}^T\mathbf{H})^{-1}\sigma^2 \tag{9.64}$$

and is a measure of how well we can estimate the unknown $\mathbf{x}$. The square root of the diagonal elements of $\mathbf{P}$ represent the rms values of the errors in each element in $\mathbf{x}$.

◆ **Example 9.7** *Least-Squares Fit*

The monthly sales (in thousands \$) for the first year of the Mewisham Co. are given by

$$\mathbf{y}^T = [0.2\ 0.5\ 1.1\ 1.2\ 1.1\ 1.3\ 1.1\ 1.2\ 2.0\ 1.2\ 2.2\ 4.0].$$

Find the least-squares fit parabola to this data and use that to predict what the monthly sales will be during the second year. Also state what the predicted accuracy of the parabolic coefficients are, assuming that the rms accuracy of the data is \$700.

Solution. The solution is obtained using Eq. (9.60), where $\mathbf{H}$ contains the parabolic function to be fit. Each month's sales obeys

$$y_i = a_o + a_1t_i + a_2t_i^2 + v_i,$$

7 $\mathcal{E}$ is called the **expectation** and effectively means the average of the quantity in { }.

so that, in vector form

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 & t_1 & t_1^2 \\ 1 & t_2 & t_2^2 \\ 1 & t_3 & t_3^2 \\ \vdots & \vdots & \vdots \end{bmatrix} \begin{bmatrix} a_o \\ a_1 \\ a_2 \end{bmatrix} + \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \end{bmatrix},$$

and we see that

$$\mathbf{H} = \begin{bmatrix} 1 & t_1 & t_1^2 \\ 1 & t_2 & t_2^2 \\ 1 & t_3 & t_3^2 \\ \vdots & \vdots & \vdots \end{bmatrix} \qquad t_i = i, \quad i = 1, 12$$

and

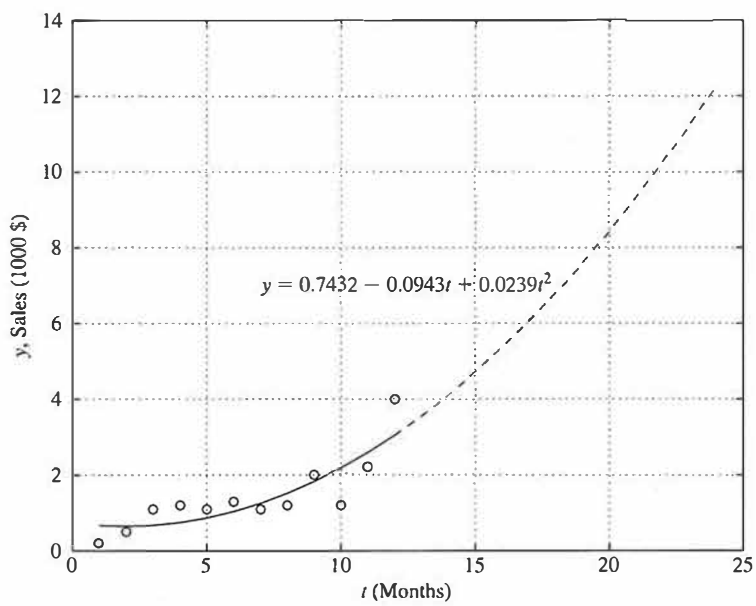
$$\mathbf{x} = \begin{bmatrix} a_o \\ a_1 \\ a_2 \end{bmatrix}.$$

Evaluation of Eq. (9.60) produces an estimate of $\mathbf{x}$

$$\hat{x} = \begin{bmatrix} 0.7432 \\ -0.0943 \\ 0.0239 \end{bmatrix}$$

which is used to plot the "best fit" parabola along with the raw data in Fig. 9.7. The data used to determine the parabola only occurred during the first 12 months, after that the parabola is an extrapolation.

Figure 9.7
Least-squares fit of parabola to data in Example 9.7



Equation (9.64), with $\mathbf{H}$ as above and $\sigma^2 = 0.49$ (σ was given to be 0.7), shows that

$$\mathbf{P} = \begin{bmatrix} 0.5234 & -0.1670 & 0.0111 \\ -0.1670 & 0.0655 & -0.0048 \\ 0.0111 & -0.0048 & 0.0004 \end{bmatrix},$$

which means that the rms accuracy of the coefficients are

$$\sigma_{a_o} = \sqrt{0.5234} = 0.7235$$
$$\sigma_{a_1} = \sqrt{0.0655} = 0.2559$$
$$\sigma_{a_2} = \sqrt{0.0004} = 0.0192.$$

◆

Weighted Least Squares

In many cases, we know *a priori* that some measurements are more accurate than others so that all the diagonal elements in $\mathbf{R}$ are *not* the same. In this case, it makes sense to weight the measurement errors higher for those measurements known to be more accurate because that will cause those measurements to have a bigger influence on the cost minimization. In other words, the cost function in Eq. (9.58) needs to be modified to

$$\mathcal{J} = \frac{1}{2}\mathbf{v}^T\mathbf{W}\mathbf{v}, \tag{9.65}$$

where $\mathbf{W}$ is a diagonal weighting matrix whose elements are in some way inversely related to the uncertainty of the corresponding element of $\mathbf{v}$. Performing the same algebra as for the unweighted case above, we find that the best **weighted least squares** solution is given by

$$\hat{\mathbf{x}} = [\mathbf{H}^T\mathbf{W}\mathbf{H}]^{-1}\mathbf{H}^T\mathbf{W}\mathbf{y}. \tag{9.66}$$

The covariance of this estimate also directly follows the development of Eq. (9.62) and results in

$$\mathbf{P} = (\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}\mathbf{H}^T\mathbf{W}\mathcal{E}\{\mathbf{v}\mathbf{v}^T\}\mathbf{W}\mathbf{H}(\mathbf{H}^T\mathbf{W}\mathbf{H})^{-1}. \tag{9.67}$$

A logical choice for $\mathbf{W}$ is to let it be inversely proportional to $\mathbf{R}$, thus weighting the square of the measurement errors exactly in proportion to the inverse of their *a priori* mean square error, that is, let

$$\mathbf{W} = \mathbf{R}^{-1}. \tag{9.68}$$

This choice of weighting matrix is proven in Section 12.7 to minimize the trace of $\mathbf{P}$ and is called the **best linear unbiased estimate**. With this choice, Eq. (9.66) becomes

$$\hat{\mathbf{x}} = [\mathbf{H}^T\mathbf{R}^{-1}\mathbf{H}]^{-1}\mathbf{H}^T\mathbf{R}^{-1}\mathbf{y}, \tag{9.69}$$

and Eq. (9.67) reduces to

$$\mathbf{P} = (\mathbf{H}^T\mathbf{R}^{-1}\mathbf{H})^{-1}. \tag{9.70}$$

Recursive Least Squares

The two least-squares algorithms above, Eqs. (9.60) and (9.69), are both **batch** algorithms in that all the data is obtained and then processed in one calculation. For an estimation problem that runs for a long time, the measurement vector would become very large, and one would have to wait until the problem was complete in order to calculate the estimate. The recursive formulation solves both these difficulties by performing the calculation in small time steps. The ideas are precisely the same, that is, a weighted least squares calculation is being performed. But now we break the problem into old data, for which we already found $\hat{\mathbf{x}}$, and new data, for which we want a correction to $\hat{\mathbf{x}}$, so that the overall new $\hat{\mathbf{x}}$, is adjusted for the newly acquired data.

The problem is stated as

$$\begin{bmatrix} \mathbf{y}_o \\ \mathbf{y}_n \end{bmatrix} = \begin{bmatrix} \mathbf{H}_o \\ \mathbf{H}_n \end{bmatrix} \mathbf{x} + \begin{bmatrix} \mathbf{v}_o \\ \mathbf{v}_n \end{bmatrix}, \tag{9.71}$$

where the subscript o represents old data and n represents new data. The best estimate of $\mathbf{x}$ given all the data follows directly from Eq. (9.69) and can be written as

$$\begin{bmatrix} \mathbf{H}_o \\ \mathbf{H}_n \end{bmatrix}^T \begin{bmatrix} \mathbf{R}_o^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_n^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{H}_o \\ \mathbf{H}_n \end{bmatrix} \hat{\mathbf{x}} = \begin{bmatrix} \mathbf{H}_o \\ \mathbf{H}_n \end{bmatrix}^T \begin{bmatrix} \mathbf{R}_o^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_n^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{y}_o \\ \mathbf{y}_n \end{bmatrix}, \tag{9.72}$$

where $\hat{\mathbf{x}}$ is the best estimate of $\mathbf{x}$ given all the data, old and new. Let's define $\hat{\mathbf{x}}$ as

$$\hat{\mathbf{x}} = \hat{\mathbf{x}}_o + \delta\hat{\mathbf{x}}, \tag{9.73}$$

where $\mathbf{x}_o$ is the best estimate given only the old data, $\mathbf{y}_o$. We want to find an expression for the correction to this estimate, $\delta\hat{\mathbf{x}}$, given the new data. Since $\mathbf{x}_o$ was the best estimate given the old data, it satisfies

$$[\mathbf{H}_o^T\mathbf{R}_o^{-1}\mathbf{H}_o]\hat{\mathbf{x}}_o = \mathbf{H}_o^T\mathbf{R}_o^{-1}\mathbf{y}_o. \tag{9.74}$$

Expanding out the terms in Eq. (9.72) and using Eqs. (9.73) and (9.74) yields

$$\mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n\hat{\mathbf{x}}_o + [\mathbf{H}_o^T\mathbf{R}_o^{-1}\mathbf{H}_o + \mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]\delta\hat{\mathbf{x}} = \mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{y}_n, \tag{9.75}$$

which can be solved for the desired result

$$\delta\hat{\mathbf{x}} = [\mathbf{H}_o^T\mathbf{R}_o^{-1}\mathbf{H}_o + \mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]^{-1}\mathbf{H}_n^T\mathbf{R}_n^{-1}(\mathbf{y}_n - \mathbf{H}_n\mathbf{x}_o). \tag{9.76}$$

Equation (9.70) defined the covariance of the estimate, and in terms of the old data would be written as

$$\mathbf{P}_o = (\mathbf{H}_o^T\mathbf{R}_o^{-1}\mathbf{H}_o)^{-1}, \tag{9.77}$$

so that Eq. (9.76) reduces to

$$\delta\hat{\mathbf{x}} = [\mathbf{P}_o^{-1} + \mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]^{-1}\mathbf{H}_n^T\mathbf{R}_n^{-1}(\mathbf{y}_n - \mathbf{H}_n\hat{\mathbf{x}}_o), \tag{9.78}$$

and we see that the correction to the old estimate can be determined if we simply know the old estimate and its covariance. Note that the correction to $\mathbf{x}$ is proportional to the *difference* between the new data $\mathbf{y}_n$ and the estimate of the new data $\mathbf{H}_n\hat{\mathbf{x}}_o$ based on the old $\mathbf{x}$.

By analogy with the weighted least squares, the covariance of the new estimate is

$$\mathbf{P}_n = [\mathbf{P}_o^{-1} + \mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]^{-1} \tag{9.79}$$

Note here that it is no longer necessary for there to be more new measurements than the elements in $\mathbf{x}$. The only requirement is that $\mathbf{P}_n$ be full rank, which could be satisfied by virtue of $\mathbf{P}_o$ being full rank. In other words, in Example 9.7 it would have been possible to start the process with the first three months of sales, then recursively update the parabolic coefficients using one month's additional sales at a time. In fact, we will see in the Kalman filter that it is typical for the new $\mathbf{y}$ to have fewer elements than $\mathbf{x}$.

recursive least-square procedure

To summarize the procedure, we start by assuming that $\mathbf{x}_o$ and $\mathbf{P}_o$ are available from previous calculations.

- Compute the new covariance from Eq. (9.79)

$$\mathbf{P}_n = [\mathbf{P}_o^{-1} + \mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]^{-1}.$$

- Compute the new value of $\hat{\mathbf{x}}$ using Eqs. (9.73), (9.78), and (9.79)

$$\hat{\mathbf{x}} = \hat{\mathbf{x}}_o + \mathbf{P}_n\mathbf{H}_n^T\mathbf{R}_n^{-1}(\mathbf{y}_n - \mathbf{H}_n\hat{\mathbf{x}}_o). \tag{9.80}$$

- Take new data and repeat the process.

This algorithm assigns relative weighting to the old $\hat{\mathbf{x}}$ vs. new data based on their relative accuracies, similarly to the weighted least squares. For example, if the old data produced an extremely accurate estimate so that $\mathbf{P}_o$ was almost zero, then Eq. (9.79) shows that $\mathbf{P}_n$ is $\cong 0$ and Eq. (9.80) shows that the new estimate will essentially ignore the new data. On the other hand, if the old estimates are very poor, that is, $\mathbf{P}_o$ is very large, Eq. (9.79) shows that

$$\mathbf{P}_n \cong [\mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]^{-1},$$

and Eq. (9.80) shows that

$$\hat{\mathbf{x}} \cong [\mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{H}_n]^{-1}\mathbf{H}_n^T\mathbf{R}_n^{-1}\mathbf{y}_n,$$

which ignores the old data. Most cases are somewhere between these two extremes, but the fact remains that this recursive-weighted-least-squares algorithm weights the old and new data according to the associated covariance.

9.4.2 The Kalman Filter

Now consider a discrete dynamic plant

$$\mathbf{x}(k+1) = \mathbf{\Phi}\mathbf{x}(k) + \mathbf{\Gamma}\mathbf{u}(k) + \mathbf{\Gamma}_1\mathbf{w}(k) \tag{9.81}$$

with measurements

$$\mathbf{y}(k) = \mathbf{H}\mathbf{x}(k) + \mathbf{v}(k), \tag{9.82}$$

where the **process noise** $\mathbf{w}(k)$ and **measurement noise** $\mathbf{v}(k)$ are random sequences with zero mean, that is

$$\mathcal{E}\{\mathbf{w}(k)\} = \mathcal{E}\{\mathbf{v}(k)\} = \mathbf{0},$$

have no time correlation or are "white" noise, that is

$$\mathcal{E}\{\mathbf{w}(i)\mathbf{w}^T(j)\} = \mathcal{E}\{\mathbf{v}(i)\mathbf{v}^T(j)\} = 0 \qquad \text{if } i \neq j,$$

and have covariances or mean square "noise levels" defined by

$$\mathcal{E}\{\mathbf{w}(k)\mathbf{w}^T(k)\} = \mathbf{R}_w, \qquad \mathcal{E}\{\mathbf{v}(k)\mathbf{v}^T(k)\} = \mathbf{R}_v.$$

We allow $\mathbf{L}$ in Eq. (9.56) to vary with the time step, and we wish to pick $\mathbf{L}(k)$ so that the estimate of $\mathbf{x}(k)$, given all the data up to and including time k, is optimal.

Let us pretend temporarily that without using the current measurement $\mathbf{y}(k)$, we already have a prior estimate of the state at the time of a measurement, which we will call $\bar{\mathbf{x}}(k)$. The problem at this point is to update this old estimate based on the current new measurement.

Comparing this problem to the recursive least squares, we see that the estimation measurement equation Eq. (9.82) relates the new measurements to $\mathbf{x}$ just as the lower row in Eq. (9.71) does; hence the optimal state estimation solution is given by Eq. (9.80), where $\mathbf{x}_o$ takes the role of $\bar{\mathbf{x}}(k)$, $\mathbf{P}_n = \mathbf{P}(k)$, $\mathbf{H}_n = \mathbf{H}$, and $\mathbf{R}_n = \mathbf{R}_v$. The solution equations are

$$\hat{\mathbf{x}}(k) = \bar{\mathbf{x}}(k) + \mathbf{L}(k)(\mathbf{y}(k) - \mathbf{H}\bar{\mathbf{x}}(k)), \tag{9.83}$$

where,

$$\mathbf{L}(k) = \mathbf{P}(k)\mathbf{H}^T\mathbf{R}_v^{-1}. \tag{9.84}$$

Equation (9.79) is used to find $\mathbf{P}(k)$, where we now call the old covariance $\mathbf{M}(k)$ instead of $\mathbf{P}_o$, thus

$$\mathbf{P}(k) = [\mathbf{M}^{-1} + \mathbf{H}^T\mathbf{R}_v^{-1}\mathbf{H}]^{-1}. \tag{9.85}$$

The size of the matrix to be inverted in Eq. (9.85) is $n \times n$, where n is the dimension of $\mathbf{x}$. For the Kalman filter, $\mathbf{y}$ usually has fewer elements than $\mathbf{x}$, and it is more efficient to use the matrix inversion lemma (See Eq. (C.6) in Appendix C) to convert Eq. (9.85) to

$$\mathbf{P}(k) = \mathbf{M}(k) - \mathbf{M}(k)\mathbf{H}^T(\mathbf{H}\mathbf{M}(k)\mathbf{H}^T + \mathbf{R}_v)^{-1}\mathbf{H}\mathbf{M}(k). \tag{9.86}$$

$\mathbf{M}(k)$ is the covariance (or expected mean square error) of the state estimate, $\bar{\mathbf{x}}(k)$, before the measurement. The state estimate after the measurement, $\hat{\mathbf{x}}(k)$, has an error covariance $\mathbf{P}(k)$.

The idea of combining the previous estimate with the current measurement based on the relative accuracy of the two quantities—the recursive least-squares concept—was the genesis for the relationships in Eq. (9.83) and Eq. (9.85) and is one of the basic ideas of the Kalman filter. The other key idea has to do with using the known dynamics of $\mathbf{x}$ to predict its behavior between samples, an idea that was discussed in the development of the estimator in Chapter 8.

Use of dynamics in the propagation of the estimate of $\mathbf{x}$ between $k-1$ and k did not come up in the static least-squares estimation, but here this issue needs to be addressed. The state estimate at $k-1$, given data up through $k-1$, is called $\hat{\mathbf{x}}(k-1)$, whereas we defined $\bar{\mathbf{x}}(k)$ to be the estimate at k given the same data up through $k-1$. In the static least squares, these two quantities were identical and both called $\mathbf{x}_o$ because it was presumed to be a constant; but here the estimates differ due to the fact that the state will change according to the system dynamics as time passes. Specifically, the estimate $\bar{\mathbf{x}}(k)$ is found from $\hat{\mathbf{x}}(k-1)$ by using Eq. (9.81) with $\mathbf{w}(k-1)=0$, because we know that this is the expected value of $\mathbf{x}(k)$ since the expected value of the plant noise, $\mathcal{E}\{\mathbf{w}(k-1)\}$ is zero. Thus

$$\bar{\mathbf{x}}(k) = \mathbf{\Phi}\hat{\mathbf{x}}(k-1) + \mathbf{\Gamma}\mathbf{u}(k-1). \tag{9.87}$$

The change in estimate from $\hat{\mathbf{x}}(k-1)$ to $\bar{\mathbf{x}}(k)$ is called a "time update," whereas the change in the estimate from $\bar{\mathbf{x}}(k)$ to $\hat{\mathbf{x}}(k)$ as given by Eq. (9.83) is a "measurement update," which occurs at the fixed time k but expresses the improvement in the estimate due to the measurement $\mathbf{y}(k)$. The same kind of time and measurement updates apply to the estimate covariances, $\mathbf{P}$ and $\mathbf{M}$: $\mathbf{P}$ represents the estimate accuracy immediately after a measurement, whereas $\mathbf{M}$ is the propagated value of $\mathbf{P}$ and is valid just before measurements. From Eq. (9.81) and Eq. (9.87) we see that

$$\mathbf{x}(k+1) - \bar{\mathbf{x}}(k+1) = \mathbf{\Phi}(\mathbf{x}(k) - \hat{\mathbf{x}}(k)) + \mathbf{\Gamma}_1\mathbf{w}(k), \tag{9.88}$$

which we will use to find the covariance of the state at time $k+1$ *before* taking $\mathbf{y}(k+1)$ into account

$$\mathbf{M}(k+1) = \mathcal{E}[(\mathbf{x}(k+1) - \bar{\mathbf{x}}(k+1))(\mathbf{x}(k+1) - \bar{\mathbf{x}}(k+1))^T].$$

If the measurement noise, $\mathbf{v}$, and the process noise, $\mathbf{w}$, are uncorrelated so that $\mathbf{x}(k)$ and $\mathbf{w}(k)$ are also uncorrelated, the cross product terms vanish and we find that

$$\mathbf{M}(k+1) = \mathcal{E}[\mathbf{\Phi}(\mathbf{x}(k) - \hat{\mathbf{x}}(k))(\mathbf{x}(k) - \hat{\mathbf{x}}(k))^T\mathbf{\Phi}^T + \mathbf{\Gamma}_1\mathbf{w}(k)\mathbf{w}^T(k)\mathbf{\Gamma}_1^T]. \tag{9.89}$$

But because

$$\mathbf{P}(k) = \mathcal{E}\{(\mathbf{x}(k) - \hat{\mathbf{x}}(k))(\mathbf{x}(k) - \hat{\mathbf{x}}(k))^T\} \quad \text{and} \quad \mathbf{R}_w = \mathcal{E}\{\mathbf{w}(k)\mathbf{w}^T(k)\},$$

Eq. (9.89) reduces to

$$\mathbf{M}(k+1) = \mathbf{\Phi}\mathbf{P}(k)\mathbf{\Phi}^T + \mathbf{\Gamma}_1\mathbf{R}_w\mathbf{\Gamma}_1^T. \tag{9.90}$$

This completes the required relations for the optimal, time-varying gain, state estimation, commonly referred to as the Kalman filter. A summary of the required relations is:

Kalman filter equations

- *At the measurement time* (measurement update)

$$\hat{\mathbf{x}}(k) = \bar{\mathbf{x}}(k) + \mathbf{P}(k)\mathbf{H}^T\mathbf{R}_v^{-1}(\mathbf{y}(k) - \mathbf{H}\bar{\mathbf{x}}(k)), \tag{9.91}$$

where

$$\mathbf{P}(k) = \mathbf{M}(k) - \mathbf{M}(k)\mathbf{H}^T(\mathbf{H}\mathbf{M}(k)\mathbf{H}^T + \mathbf{R}_v)^{-1}\mathbf{H}\mathbf{M}(k). \tag{9.92}$$

- *Between measurements* (time update)

$$\bar{\mathbf{x}}(k+1) = \mathbf{\Phi}\hat{\mathbf{x}}(k) + \mathbf{\Gamma}\mathbf{u}(k) \tag{9.93}$$

and

$$\mathbf{M}(k+1) = \mathbf{\Phi}\mathbf{P}(k)\mathbf{\Phi}^T + \mathbf{\Gamma}_1\mathbf{R}_w\mathbf{\Gamma}_1^T, \tag{9.94}$$

where the initial conditions for $\bar{\mathbf{x}}(0)$ and $\mathbf{M}(0) = \mathcal{E}\{\tilde{\mathbf{x}}(0)\tilde{\mathbf{x}}^T(0)\}$ must be assumed to be some value for initialization.

Because $\mathbf{M}$ is time-varying, so will be the estimator gain, $\mathbf{L}$, given by Eq. (9.84). Furthermore, we see that the structure of the estimation process is exactly the same as the current estimator given by Eq. (9.56), the difference being that $\mathbf{L}$ is time varying and determined so as to provide the minimum estimation errors, given *a priori* knowledge of the process noise magnitude, $\mathbf{R}_w$, the measurement noise magnitude, $\mathbf{R}_v$, and the covariance initial condition, $\mathbf{M}(0)$.

◆ **Example 9.8** *Time-Varying Kalman Filter Gains*

Solve for $\mathbf{L}(k)$ for the satellite attitude control problem in Example 9.3 assuming the angle, θ, is sensed with a measurement noise covariance

$$R_v = 0.1 \text{ deg}^2.$$

Assume the process noise is due to disturbance torques acting on the spacecraft with the disturbance-input distribution matrix

$$\mathbf{G}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$

and assume several values for the mean square magnitude of this disturbance:

$$\mathbf{R}_w = 0.001,\ 0.01,\ 0.1\ (\text{deg}^2/\text{sec}^4).$$

Solution. Because only θ is directly sensed, we have from Eq. (4.47)

$$H = [1 \quad 0].$$

The time varying estimator gain, $\mathbf{L}(k)$, is found by evaluating Eqs. (9.84), (9.92), and (9.94). In order to start these recursive equations, some value for $\mathbf{M}(0)$ is required. Although somewhat arbitrary, a value of

$$\mathbf{M}(0) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

was selected in order to indicate that the initial rms uncertainty in θ was 1° and $\dot{\theta}$ was 1°/sec. The gain time histories are sensitive to this initial condition estimate during the initial transient, but the steady final values are not affected.

Figure 9.8 shows the time history of the $\mathbf{L}$'s for the values given above. We see from the figure that after an initial settling time, the estimator gains essentially reach a steady state. Just as for the control problem, where the feedback gain reached a steady value early in the problem time, the eventual steady value for $\mathbf{L}$ occurs for all linear constant coefficient systems. The subject of the next section is a method to compute the value of this steady-state gain matrix so that it can be used in place of the time-varying one, thus eliminating the need to go through the rather lengthy recursive computation of Eq. (9.92) and Eq. (9.94).

Given an actual design problem, one can often assign a meaningful value to $\mathbf{R}_v$, which is based on the sensor accuracy. The same cannot be said for $\mathbf{R}_w$. The

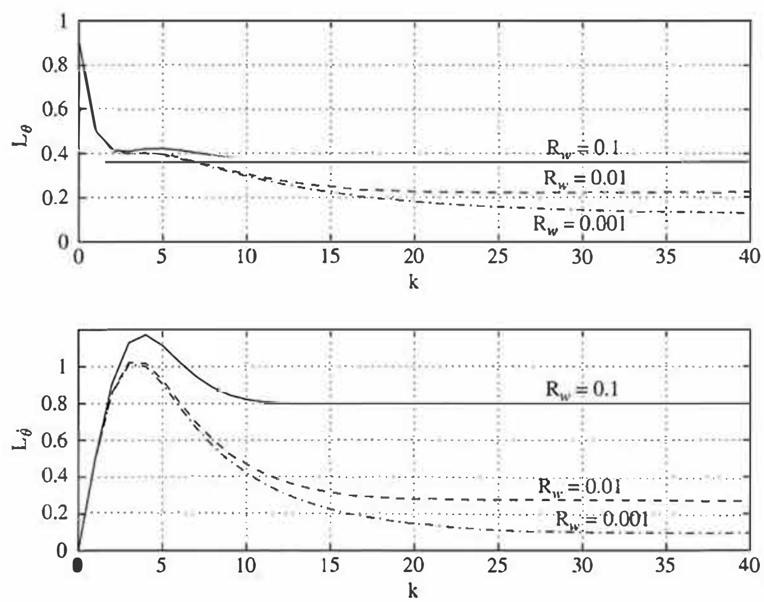


Figure 9.8
Example of estimator gains versus time

assumption of white process noise is often a mathematical artifice that is used because of the ease of solving the resulting optimization problem. Physically, $\mathbf{R}_w$ is crudely accounting for unknown disturbances, whether they be steps, white noise, or somewhere in between, and for imperfections in the plant model.

If there is a random disturbance that is time correlated—that is, **colored noise**—it can be accurately modeled by augmenting $\boldsymbol{\Phi}$ with a coloring filter that converts a white-noise input into time-correlated noise, thus Eq. (9.81) can also be made to describe nonwhite disturbances. In practice, however, this is often not done due to the complexity. Instead, the disturbances are assumed white, and the noise intensity is adjusted to give acceptable results in the presence of expected disturbances, whether time-correlated or not.

If $\mathbf{R}_w$ was chosen to be zero due to a lack of knowledge of a precise noise model, the estimator gain would eventually go to zero. This is so because the optimal thing to do in the idealistic situation of *no* disturbances and a *perfect* plant model is to estimate open loop after the initial condition errors have *completely* died out; after all, the very best filter for the noisy measurements is to totally ignore them. In practice, this will not work because there are always some disturbances and the plant model is never perfect; thus, the filter with zero gain will drift away from reality and is referred to as a **divergent filter**.[8] If an estimator mode with zero gain was also naturally unstable, the estimator error would diverge from reality very quickly and likely result in saturation of the computer. We therefore are often forced to pick values of $\mathbf{R}_w$ and sometimes $\boldsymbol{\Gamma}_1$ "out of a hat" in the design process in order to assure that no modes of the estimator are without feedback and that the estimator will track all modes of the actual system. The disturbance noise model should be selected to approximate that of the actual known disturbances when practical, but the designer often settles on acceptable values based on the quality of the estimation that results in subsequent simulations including all known disturbances, white and otherwise.

divergent filters

It is possible to include a nonzero process noise input in the plant model and yet still end up with a divergent filter [Bryson (1978)]. This can arise when the process noise is modeled so that it does not affect some of the modes of the system. Bryson showed that the filter will not be divergent if you select $\boldsymbol{\Gamma}_1$ so that the system, $(\boldsymbol{\Phi}, \boldsymbol{\Gamma}_1)$, is *controllable* and all diagonal elements of $\mathbf{R}_w$ are nonzero.

For an implementation of a time-varying filter, initial conditions for $\mathbf{M}$ and $\bar{\mathbf{x}}$ are also required. Physically, they represent the *a priori* estimate of the accuracy of $\bar{\mathbf{x}}(0)$, which in turn represents the *a priori* estimate of the state. In some cases, there might be test data of some kind to support their intelligent choice; however, that is not typical. In lieu of any better information, one could logically assume that the components of $\bar{\mathbf{x}}(0)$ contained in $\mathbf{y}$ are equal to the first measurement, and the remaining components are equal to zero. Similarly, the components in $\mathbf{M}$ that

8 A filter with one or more modes that is running open loop is sometimes dubbed "oblivious" or "fat, dumb, and happy" because it ignores the measurements. It is somewhat analogous to the person whose mind is made up and not interested in facts.

represent a measured component could be logically set to $\mathbf{R}_v$ and the remaining components set to a high value.

In obtaining the values of $\mathbf{L}$ in Fig. 9.8, it was necessary to solve for the time history of $\mathbf{P}$. Because $\mathbf{P}$ is the covariance of the estimation errors, we can sometimes use $\mathbf{P}$ as an indicator of estimation accuracy, provided that the values of $\mathbf{R}_v$ and $\mathbf{R}_w$ are based on some knowledge of the actual noise characteristics and that $\mathbf{w}$ and $\mathbf{v}$ are approximately white.

The intent of this section and example is to give some insight into the nature of the solution so as to motivate and provide a basis for the following section. Readers interested in the application of Kalman filters are encouraged to review works devoted to that subject, such as Bryson and Ho (1975), Anderson and Moore (1979), and Stengel (1986).

9.4.3 Steady-State Optimal Estimation

As shown by Example 9.8, the estimator gains will eventually reach a steady-state value if enough time passes. This is so because the values of $\mathbf{M}$ and $\mathbf{P}$ reach a steady value. Because of the substantial simplification in the controller afforded by a constant estimator gain matrix, it is often desirable to determine the constant gain during the design process and to implement that constant value in the controller. As discussed in Section 9.2, many control systems run for very long times and can be treated mathematically as if they run for an infinite time. In this case the constant gain is the optimal because the early transient period has no significant effect. Whatever the motivation, a constant-gain Kalman filter is identical in structure to the estimator discussed in Chapter 8, the only difference being that the gain, $\mathbf{L}$, is determined so that the estimate errors are minimized for the assumed level of process and measurement noise. This approach replaces the pole-placement method of finding the estimator gain and has the highly desirable feature that it can be applied to MIMO systems.

The equations to be solved that determine $\mathbf{M}$ and $\mathbf{P}$ are Eq. (9.92) and Eq. (9.94). Repeated, they are

$$\mathbf{P}(k) = \mathbf{M}(k) - \mathbf{M}(k)\mathbf{H}^T(\mathbf{H}\mathbf{M}(k)\mathbf{H}^T + \mathbf{R}_v)^{-1}\mathbf{H}\mathbf{M}(k), \tag{9.92}$$

$$\mathbf{M}(k+1) = \mathbf{\Phi}\mathbf{P}(k)\mathbf{\Phi}^T + \mathbf{\Gamma}_1\mathbf{R}_w\mathbf{\Gamma}_1^T \tag{9.94}$$

Comparing Eqs. (9.92) and (9.94) to the optimal *control* recursion relationships, Eq. (9.24) and Eq. (9.25)

$$\mathbf{M}(k) = \mathbf{S}(k) - \mathbf{S}(k)\mathbf{\Gamma}[\mathbf{Q}_2 + \mathbf{\Gamma}^T\mathbf{S}(k)\mathbf{\Gamma}]^{-1}\mathbf{\Gamma}^T\mathbf{S}(k), \tag{9.25}$$

$$\mathbf{S}(k) = \mathbf{\Phi}^T\mathbf{M}(k+1)\mathbf{\Phi} + \mathbf{Q}_1, \tag{9.24}$$

duality

we see that they are precisely of the same form! The only exception is that Eq. (9.94) goes forward instead of backward as Eq. (9.24) does. Therefore, we can simply change variables and directly use the steady-state solution of the control problem as the desired steady-state solution to the estimation problem, even though the equations are solved in opposite directions in time.

Table 9.1 lists the correspondences that result by direct comparison of the control and estimation recursion relations: Eq. (9.25) with Eq. (9.92) and Eq. (9.24) with Eq. (9.94).

By analogy with the control problem, Eqs. (9.92) and (9.94) must have arisen from two coupled equations with the same form as Eq. (9.33). Using the correspondences in Table 9.1, the control Hamiltonian in Eq. (9.34) becomes the estimation Hamiltonian

$$\mathcal{H}_e = \begin{bmatrix} \boldsymbol{\Phi}^T + \mathbf{H}^T\mathbf{R}_v^{-1}\mathbf{H}\boldsymbol{\Phi}^{-1}\boldsymbol{\Gamma}_1\mathbf{R}_w\boldsymbol{\Gamma}_1^T & -\mathbf{H}^T\mathbf{R}_v^{-1}\mathbf{H}\boldsymbol{\Phi}^{-1} \\ -\boldsymbol{\Phi}^{-1}\boldsymbol{\Gamma}_1\mathbf{R}_w\boldsymbol{\Gamma}_1^T & \boldsymbol{\Phi}^{-1} \end{bmatrix}. \tag{9.95}$$

Therefore, the steady-state value of $\mathbf{M}$ is deduced by comparison with Eq. (9.46) and is

$$\mathbf{M}_\infty = \boldsymbol{\Lambda}_I\mathbf{X}_I^{-1}, \tag{9.96}$$

where

$$\begin{bmatrix} \mathbf{X}_I \\ \boldsymbol{\Lambda}_I \end{bmatrix}$$

are the eigenvectors of $\mathcal{H}_e$ associated with its stable eigenvalues. Hence, from Eqs. (9.92) and (9.84) after some manipulation we find the steady state Kalman-filter gain to be[9]

$$\mathbf{L}_\infty = \mathbf{M}_\infty\mathbf{H}^T(\mathbf{H}\mathbf{M}_\infty\mathbf{H}^T + \mathbf{R}_v)^{-1}. \tag{9.97}$$

This is a standard calculation in MATLAB's kalman.m. Sometimes this solution is referred to as the **linear quadratic Gaussian** (LQG) problem because it is often assumed in the derivation that the noise has a Gaussian distribution. As can be seen from the development here, this assumption is not necessary. However, with this assumption, one can show that the estimate is not only the one that minimizes

LQG

Table 9.1 Control and Estimation Duality

Control	*Estimation*
$\boldsymbol{\Phi}$	$\boldsymbol{\Phi}^T$
$\mathbf{M}$	$\mathbf{P}$
$\mathbf{S}$	$\mathbf{M}$
$\mathbf{Q}_1$	$\boldsymbol{\Gamma}_1\mathbf{R}_w\boldsymbol{\Gamma}_1^T$
$\boldsymbol{\Gamma}$	$\mathbf{H}^T$
$\mathbf{Q}_2$	$\mathbf{R}_v$

9 In the steady state, the filter has constant coefficients and, for the assumed model, is the same as the Wiener filter.

the error squared, but also the one that is the statistically "most likely." In this derivation, the result is referred to as the **maximum likelihood** estimate.

Because Eq. (9.34) and Eq. (9.95) are the same form, the eigenvalues have the same reciprocal properties in both cases. Furthermore, for systems with a single output and a single process noise input, the symmetric root locus follows by analogy with Eq. (9.40) and the use of Table 9.1. Specifically, the characteristic equation becomes

$$1 + qG_e(z^{-1})G_e(z) = 0, \tag{9.98}$$

where $q = R_w / R_v$ and

$$G_e(z) = \mathbf{H}(z\mathbf{I} - \mathbf{\Phi})^{-1}\mathbf{\Gamma}_1.$$

Therefore, for systems where the process noise is additive with the control input—that is, $\mathbf{\Gamma} = \mathbf{\Gamma}_1$—the control and estimation optimal root loci are identical, and the control and estimation roots could be selected from the same loci. For example, the loci in Figs. 9.4 and 9.5 could be used to select estimator roots as well as the control roots.

9.4.4 Noise Matrices and Discrete Equivalents

The quantities defining the noise magnitude, the covariance matrices $\mathbf{R}_w$ and $\mathbf{R}_v$, were defined in Section 9.4.2 as

$$\mathbf{R}_w = \mathcal{E}\{\mathbf{w}(k)\mathbf{w}^T(k)\} \quad \text{and} \quad \mathbf{R}_v = \mathcal{E}\{\mathbf{v}(k)\mathbf{v}^T(k)\}.$$

Typically, if there is more than one process or measurement noise component, one has no information on the cross-correlation of the noise elements and therefore $\mathbf{R}_w$ and $\mathbf{R}_v$ are selected as diagonal matrices. The magnitudes of the diagonal elements are the variances of the noise components.

Process Noise, $\mathbf{R}_w$

The process noise acts on the continuous portion of the system and, assuming that it is a white continuous process, varies widely throughout one sample period. Its effect over the sample period, therefore, cannot be determined as it was for Eq. (9.90); instead, it needs to be integrated. From Eqs. (4.55) and (4.56), we see that the effect of continuous noise input over one sample period is

$$\mathbf{x}(k+1) = \mathbf{\Phi}\mathbf{x}(k) + \int_0^T e^{\mathbf{F}(\eta)}\mathbf{G}_1\mathbf{w}(\eta)d\eta,$$

where $\mathbf{G}_1$ is defined by Eq. (4.45). Repeating the derivation[10] of Eq. (9.90) with the integral above replacing $\Gamma_1\mathbf{w}(k)$ in Eq. (9.88), we find that Eq. (9.90) becomes

$$\mathbf{M}(k+1) = \mathbf{\Phi} P(k)\mathbf{\Phi}^T + \int_0^T \int_0^T \mathbf{\Phi}(\eta)\mathbf{G}_1 \mathcal{E}[\mathbf{w}(\eta)\mathbf{w}^T(\tau)]\mathbf{G}_1^T \mathbf{\Phi}^T(\tau)d\tau d\eta. \tag{9.99}$$

But the white noise model for $\mathbf{w}$ means that

$$\mathcal{E}[\mathbf{w}(\eta)\mathbf{w}^T(\tau)] = \mathbf{R}_{\text{wpsd}}\,\delta(\eta - \tau),$$

where $\mathbf{R}_{\text{wpsd}}$ is called the power spectral density, or mean-square spectral density, of the continuous white noise. Therefore, Eq. (9.99) reduces to

$$\mathbf{M}(k+1) = \mathbf{\Phi} P(k)\mathbf{\Phi}^T + \mathbf{C}_d,$$

where

$$\mathbf{C}_d = \int_0^T \mathbf{\Phi}(\tau)\mathbf{G}_1\mathbf{R}_{\text{wpsd}}\mathbf{G}_1^T\mathbf{\Phi}^T(\tau)d\tau. \tag{9.100}$$

Calculation of this integral (see disrw.m in the Digital Control Toolbox) can be carried out using a similar exponential form due to Van Loan (1978), as in Eq. (9.54). If T is very short compared to the system time constants, that is

$$\mathbf{\Phi} \cong \mathbf{I} \quad \text{and} \quad \mathbf{\Gamma}_1 \cong \mathbf{G}_1 T,$$

then the integral is approximately

$$\mathbf{C}_d \cong T\,\mathbf{G}_1\mathbf{R}_{\text{wpsd}}\mathbf{G}_1^T,$$

which can also be written[11]

$$\mathbf{C}_d \cong \mathbf{\Gamma}_1 \frac{\mathbf{R}_{\text{wpsd}}}{T}\mathbf{\Gamma}_1^T.$$

Therefore, one can apply the discrete covariance update Eq. (9.94) to the case where $\mathbf{w}$ is continuous noise by using the approximation that

$$\mathbf{R}_w \cong \frac{\mathbf{R}_{\text{wpsd}}}{T}, \tag{9.101}$$

as long as T is much shorter than the system time constants. If this assumption is not valid, one must revert to the integral in Eq. (9.100).

In reality, however, there is no such thing as white continuous noise. A pure white-noise disturbance would have equal magnitude content at all frequencies from 0 to ∞. Translated, that means that the correlation time of the random signal is precisely zero. The only requirement for our use of the white-noise

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10 See Stengel (1986), p. 327.

11 By computing $\mathbf{\Gamma}_1$ exactly according to Eq. (4.58), the approximation that follows for $\mathbf{C}_d$ is substantially more accurate than the approximation using $\mathbf{G}_1$.

model in discrete systems is that the disturbance have a correlation time that is short compared to the sample period. If the correlation time is on the same order or longer than the sample period, the correct methodology entails adding the colored-noise model to the plant model and estimating the random disturbance along with the original state. In fact, if the correlation time is extremely long, the disturbance acts much like a bias, and we have already discussed its estimation in Section 8.5.2. In practice, disturbances are often assumed to be either white or a bias, because the improved accuracy possible by modeling a random disturbance with a time constant on the same order as the sample period is not deemed worth the extra complexity.

The determination of the appropriate value of $\mathbf{R}_{\text{wpsd}}$ that represents a physical process is aided by the realization that pure white noise does not exist in nature. Disturbances all have a nonzero correlation time, the only question being: How short? Assuming that the time correlation is exponential with a correlation time τ_c, and that $\tau_c \ll$ system time constants, the relation between the power spectral density and the mean square of the signal is[12]

$$R_{\text{wpsd}} \cong 2\tau_c \mathcal{E}\{w^2(t)\}. \tag{9.102}$$

Typically, one can measure the mean square value, $\mathcal{E}\{w^2(t)\}$, and can either measure or estimate its correlation time, τ_c, thus allowing the computation of each diagonal element of $\mathbf{R}_{\text{wpsd}}$. However, the desired result for our purposes is the discrete equivalent noise, $\mathbf{R}_{\text{w}}$. It can be computed from Eq. (9.101) and Eq. (9.102), where each diagonal element, $[\mathbf{R}_{\text{w}}]_i$, is related to the mean square and correlation time of the ith disturbance according to

$$[\mathbf{R}_{\text{w}}]_i = \frac{2}{T}[\tau_c \mathcal{E}\{w^2(t)\}]_i. \tag{9.103}$$

Note from Eq. (9.103) that $\mathbf{R}_{\text{w}}$ and $\mathcal{E}\{w^2(t)\}$ are not the same quantities. Specifically, the diagonal elements of $\mathbf{R}_{\text{w}}$ are the mean square values of the discrete noise, $w(k)$, that produces a response from the discrete model given by Eq. (9.90) that matches the response of the continuous system acted on by a $w(t)$ with $\mathcal{E}\{w^2(t)\}$ and τ_c. Note also that it has been assumed that the noise is white compared to the sample period, that is, $\tau_c \ll T$ and that $T \ll$ any system time constants. Under these conditions the discrete equivalent mean square value is less than the continuous signal mean square because the continuous random signal acting on the continuous system is averaged over the sample period. If τ_c is not $\ll T$, then the noise is not white and Eq. (9.100) is not valid; thus calculation of $\mathbf{R}_{\text{w}}$ is not relevant.

12 Bryson and Ho (1975), p. 331.

Sensor Noise, $\mathbf{R}_v$

The pertinent information given by the manufacturer of a sensor product would be the rms "jitter" error level (or some other similar name), which can usually be interpreted as the random component and assumed to be white, that is, uncorrelated from one sample to the next. The rms value is simply squared to arrive at the diagonal elements of $\mathcal{E}\{v^2(t)\}$. Unlike the process noise case these values are used directly, that is

$$[\mathbf{R}_v]_i = [\mathcal{E}\{v^2(t)\}]_i. \tag{9.104}$$

The assumption of no time correlation is consistent with the development of the optimal estimator that was discussed in Section 9.4.2. If the correlation time of a sensor is longer than the sample period, the assumption is not correct, and an accurate treatment of the noise requires that its "coloring" model be included with the plant model and the measurement noise error estimated along with the rest of the state.[13] One could also ignore this complication and proceed as if the noise were white, with the knowledge that the effect of the measurement noise is in error and that skepticism is required in the interpretation of estimate error predictions based on **P**. Furthermore, one could no longer claim that the filter was optimal.

Sensor manufacturers typically list bias errors also. This component should at least be evaluated to determine the sensitivity of the system to the bias, and if the effect is not negligible, the bias should be modeled, augmented to the state, and estimated using the ideas of Section 8.5.2.

Note that neither the sample period nor the correlation time of the sensor error has an impact on $\mathbf{R}_v$ if **v** is white. Although the rms value of **v** is not affected by the sampling, sampling at a higher rate will cause more measurements to be averaged in arriving at the state estimate, and the estimator accuracy will improve with sample rate.

In some cases, the designer wishes to know how the sensor noise will affect an estimator that is implemented with analog electronics. Although this can be done in principle, in practice it is rarely done because of the low cost of digital implementations. The value of analyzing the continuous case is that the knowledge can be useful in selecting a sample rate. Furthermore, the designer is sometimes interested in creating a digital implementation whose roots match that of a continuous design, and finding the discrete equivalent noise is a method to approximate that design goal. Whatever the reason, the continuous filter can be evaluated digitally, with the appropriate value of $\mathbf{R}_v$ being the one that provides the discrete equivalent to the continuous process, the same situation that was

13 See Stengel (1986) or Bryson and Ho (1975).

examined for the process noise. Therefore, the proper relation in this special case is from Eq. (9.101) or (9.103)

$$\mathbf{R}_v = \frac{\mathbf{R}_{vpsd}}{T} \quad \text{or} \quad [\mathbf{R}_v]_i = \frac{2}{T}[\tau_c \mathcal{E}\{v^2(t)\}]_i, \tag{9.105}$$

where $\mathcal{E}\{v^2(t)\}$ is the mean square value of the sensor noise and τ_c is its correlation time. Alternatively, if one desires only the continuous filter performance for a baseline, one can use a pure continuous analysis of the filter,[14] which requires only $\mathbf{R}_{vpsd}$.

9.5 Multivariable Control Design

The elements of the design process of a MIMO system have been discussed in the preceding sections. This section discusses some of the issues in design and provides two examples of the process.

9.5.1 Selection of Weighting Matrices $\mathbf{Q}_1$ and $\mathbf{Q}_2$

As can be seen from the discussion in Sections 9.2 and 9.3, the selection of $\mathbf{Q}_1$ and $\mathbf{Q}_2$ is only weakly connected to the performance specifications, and a certain amount of trial and error is usually required with an interactive computer program before a satisfactory design results. There are, however, a few guidelines that can be employed. For example, Bryson and Ho (1975) and Kwakernaak and Sivan (1972) suggest essentially the same approach. This is to take $\mathbf{Q}_1 = \mathbf{H}^T\overline{\mathbf{Q}}_1\mathbf{H}$ so that the states enter the cost via the important outputs (which may lead to $\mathbf{H} = \mathbf{I}$ if all states are to be kept under close regulation) and to select $\overline{\mathbf{Q}}_1$ and $\mathbf{Q}_2$ to be diagonal with entries so selected that a fixed percentage change of each variable makes an equal contribution to the cost.[15] For example, suppose we have three outputs with maximum deviations m_1, m_2, and m_3. The cost for diagonal $\overline{\mathbf{Q}}_1$ is

$$\bar{q}_{11}y_1^2 + \bar{q}_{22}y_2^2 + \bar{q}_{33}y_3^2.$$

Bryson's rules

The rule is that if $y_1 = \alpha m_1$, $y_2 = \alpha m_2$, and $y_3 = \alpha m_3$, then

$$\bar{q}_{11}y_1^2 = \bar{q}_{22}y_2^2 = \bar{q}_{33}y_3^2;$$

thus

$$\bar{q}_{11}\alpha^2 m_1^2 = \bar{q}_{22}\alpha^2 m_2^2 = \bar{q}_{33}\alpha^2 m_3^2.$$

14 See Bryson and Ho (1975).

15 Kwakernaak and Sivan (1972) suggest using a percentage change from nominal values; Bryson and Ho use percentage change from the maximum values.

A satisfactory solution for elements of $\overline{\mathbf{Q}}_1$ is then [16]

$$\overline{Q}_{1,11} = 1/m_1^2, \overline{Q}_{1,22} = 1/m_2^2, \overline{Q}_{1,33} = 1/m_3^2. \tag{9.106}$$

Similarly, for $\mathbf{Q}_2$ we select a matrix with diagonal elements

$$Q_{2,11} = 1/u_{1\max}^2, Q_{2,22} = 1/u_{2\max}^2. \tag{9.107}$$

There remains a scalar ratio between the state and the control terms, which we will call ρ. Thus the total cost is

$$\mathcal{J} = \rho \mathbf{x}^T \mathbf{H}^T \overline{\mathbf{Q}}_1 \mathbf{H}\mathbf{x} + \mathbf{u}^T \mathbf{Q}_2 \mathbf{u}, \tag{9.108}$$

where $\overline{\mathbf{Q}}_1$ and $\mathbf{Q}_2$ are given by Eq. (9.106), and Eq. (9.107), and ρ is to be selected by trial and error. A computer-interactive procedure that allows examination of root locations and transient response for selected values of $\overline{\mathbf{Q}}_1$ and $\mathbf{Q}_2$ expedites this process considerably.

9.5.2 Pincer Procedure

The designer can introduce another degree of freedom into this problem by requiring that all the closed-loop poles be inside a circle of radius $1/\alpha$, where $\alpha \geq 1$. If we do this, then the magnitude of every transient in the closed loop will decay at least as fast as $1/\alpha^k$, which forms pincers around the transients and allows a degree of direct control over the settling time. We can introduce this effect in the following way.

Suppose that as a modification to the performance criterion of Eq. (9.11), we consider

$$\mathcal{J}_\alpha = \sum_{k=0}^{\infty} [\mathbf{x}^T \mathbf{Q}_1 \mathbf{x} + \mathbf{u}^T \mathbf{Q}_2 \mathbf{u}] \alpha^{2k}. \tag{9.109}$$

We can distribute the scalar term α^{2k} in Eq. (9.109) as $\alpha^k \alpha^k$ and write it as

$$\begin{aligned}
\mathcal{J}_\alpha &= \sum_{k=0}^{\infty} [(\alpha^k \mathbf{x})^T \mathbf{Q}_1 (\alpha^k \mathbf{x}) + (\alpha^k \mathbf{u})^T \mathbf{Q}_2 (\alpha^k \mathbf{u})] \\
&= \sum_{k=0}^{\infty} [\mathbf{z}^T \mathbf{Q}_1 \mathbf{z} + \mathbf{v}^T \mathbf{Q}_2 \mathbf{v}],
\end{aligned} \tag{9.110}$$

where

$$\mathbf{z}(k) = \alpha^k \mathbf{x}(k), \quad \mathbf{v}(k) = \alpha^k \mathbf{u}(k). \tag{9.111}$$

The equations in $\mathbf{z}$ and $\mathbf{v}$ are readily found. Consider

$$\mathbf{z}(k+1) = \alpha^{k+1} \mathbf{x}(k+1).$$

16 Sometimes called Bryson's rules.

From Eq. (9.10) we have the state equations for $\mathbf{x}(k+1)$, so that

$$\mathbf{z}(k+1) = \alpha^{k+1}[\mathbf{\Phi x}(k) + \mathbf{\Gamma u}(k)].$$

If we multiply through by the α^{k+1}-term, we can write this as

$$\mathbf{z}(k+1) = \alpha\mathbf{\Phi}(\alpha^k\mathbf{x}(k)) + \alpha\mathbf{\Gamma}(\alpha^k\mathbf{u}(k)),$$

but from the definitions in Eq. (9.111), this is the same as

$$\mathbf{z}(k+1) = \alpha\mathbf{\Phi z}(k) + \alpha\mathbf{\Gamma v}(k). \qquad (9.112)$$

The performance function Eq. (9.110) and the equations of motion Eq. (9.112) define a new problem in optimal control for which the solution is a control law

$$\mathbf{v} = -\mathbf{Kz},$$

which, if we work backward, is

$$\alpha^k\mathbf{u}(k) = -\mathbf{K}(\alpha^k\mathbf{x}(k))$$

or

$$\mathbf{u}(k) = -\mathbf{Kx}(k). \qquad (9.113)$$

We conclude from all this that if we use the control law Eq. (9.113) in the state equations (9.10), then a trajectory results that is optimal for the performance $\mathcal{J}_\alpha$ given by Eq. (9.109). Furthermore, the state trajectory satisfies Eq. (9.111), where $\mathbf{z}(k)$ is a stable vector so that $\mathbf{x}(k)$ must decay *at least* as fast as $1/\alpha^k$, or else $\mathbf{z}(k)$ could not be guaranteed to be stable.

To apply the pincers we need to relate the settling time to the value of α. Suppose we define settling time of x_j as that time t_s such that if $x_j(0) = 1$ and all other states are zero at $k = 0$, then the transients in x_j are less than 0.01 (1% of the maximum) for all times greater than t_s. If we approximate the transient in x_j as

$$x_j(k) \approx x_j(0)(1/\alpha)^k,$$

then when $kT = t_s$, we must have

$$x_j(kT) \le 0.01x_j(0),$$

which will be satisfied if α is such that

$$(1/\alpha)^k \le 0.01 = \frac{1}{100},$$

or

$$\alpha > 100^{1/k} = 100^{T/t_s}. \qquad (9.114)$$

In summary, application of the pincer procedure requires that the designer select the settling time, t_s, within which all states should settle to less than 1%. Equation (9.114) is then used to compute α and, according to Eq. (9.112), the revised system $\alpha\mathbf{\Phi}$ and $\alpha\mathbf{\Gamma}$ for use in an LQR computation for the feedback gain matrix $\mathbf{K}$. Use of $\mathbf{K}$ with the original system, $\mathbf{\Phi}$, will produce a response of all states that settle within the prescribed t_s.

9.5.3 Paper-Machine Design Example

As an illustration of a multivariable control using optimal control techniques, we will consider control of the paper-machine head box described in Appendix A.5. The continuous equations of motion are given by

$$\dot{\mathbf{x}} = \begin{bmatrix} -0.2 & 0.1 & 1 \\ -0.05 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{x} + \begin{bmatrix} 0 & 1 \\ 0 & 0.7 \\ 1 & 0 \end{bmatrix} \mathbf{u}. \qquad (9.115)$$

We assume the designer has the following specifications on the responses of this system:

1. The maximum sampling frequency is 5 Hz ($T = 0.2$ sec).
2. The 1% settling time to demands on x_1 (= total head) should be less than 2.4 sec (12 periods).
3. The settling time to demands on x_2(= liquid level) should be less than 8 sec (40 periods).
4. The units on the states have been selected so that the maximum permissible deviation on total head, x_1, is 2.0 units and the liquid level, x_2, is 1.0.
5. The units on control have been selected so that the maximum permissible deviation on u_1 (air control) is 5 units and that of u_2 (stock control) is 10 units.

First let us apply the pincer procedure to ensure that the settling times are met. We have asked that the settling times be 2.4 sec for x_1 and 8 sec for x_2. If, for purposes of illustration, we select the more stringent of these and in Eq. (9.114) set $t_s = 2.4$ for which $t_s/T = 12$, then

$$\alpha > 100^{1/12} = 1.47.$$

Now let us select the cost matrices. Based on the specifications and the discussion in Section 9.5.1, we can conclude that $m_1 = 2$ and $m_2 = 1$, and thus

$$\overline{\mathbf{Q}}_1 = \begin{bmatrix} 0.25 & 0 \\ 0 & 1 \end{bmatrix}.$$

Because we are interested only in x_1 and x_2, the output matrix $\mathbf{H}$ in Eq. (9.108) is

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

and

$$\mathbf{Q}_1 = \begin{bmatrix} 0.25 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Furthermore, because $u_{1\max} = 5$ and $u_{2\max} = 10$

$$\mathbf{Q}_2 = \begin{bmatrix} 0.04 & 0 \\ 0 & 0.01 \end{bmatrix}.$$

Conversion of the continuous system in Eq. (9.115) to a discrete one ($T = 0.2$ sec) yields

$$\mathbf{\Phi} = \begin{bmatrix} 0.9607 & 0.0196 & 0.1776 \\ -0.0098 & 0.9999 & -0.0009 \\ 0 & 0 & 0.8187 \end{bmatrix} \quad \text{and} \quad \mathbf{\Gamma} = \begin{bmatrix} 0.0185 & 0.1974 \\ -0.0001 & 0.1390 \\ 0.1813 & 0 \end{bmatrix}.$$

These two matrices are then multiplied by the scalar 1.47 ($= \alpha$) and used in MATLAB's dlqr.m with the preceding $\mathbf{Q}_1$ and $\mathbf{Q}_2$.

The LQR calculation gives a control gain

$$\mathbf{K} = \begin{bmatrix} 6.81 & -9.79 & 3.79 \\ 0.95 & 4.94 & 0.10 \end{bmatrix},$$

and closed-loop poles at

$$z = 0.108,\ 0.491 \pm j0.068,$$

which are well within $1/1.47 = 0.68$. Fig. 9.9(a) shows the transient response to unit-state initial conditions on x_1, and Fig. 9.9(b) shows the same for initial conditions on x_2. Examination of these results shows that the requirement on settling time has been substantially exceeded because x_1 settles to within 1% by 1.8 sec, and x_2 settles within 1% by 1.6 sec for both cases. However, the control effort on u_1 is larger than its specification for both initial condition cases, and further iteration on the design is required. To correct this situation, the designer should be led by the fast response to relax the demands on response time in order to lower the overall need for control. Figure 9.10 shows the response with t_s in Eq. (9.114) selected to be 5 sec for the case with $\mathbf{x}(0) = [010]^T$. The control u_1 is just within the specification, and the response time of x_1 is 2.3 sec and that of x_2 is 2.0 sec. All specifications are now met.

It is possible to improve the design still further by noting that the response of x_2 beats its specified maximum value and settling time by a substantial margin. To capitalize on this observation, let us try relaxing the cost on x_2. After some iteration (see Problem 9.7), we find that *no* cost on x_2, that is

$$\mathbf{Q}_1 = \begin{bmatrix} 0.25 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix},$$

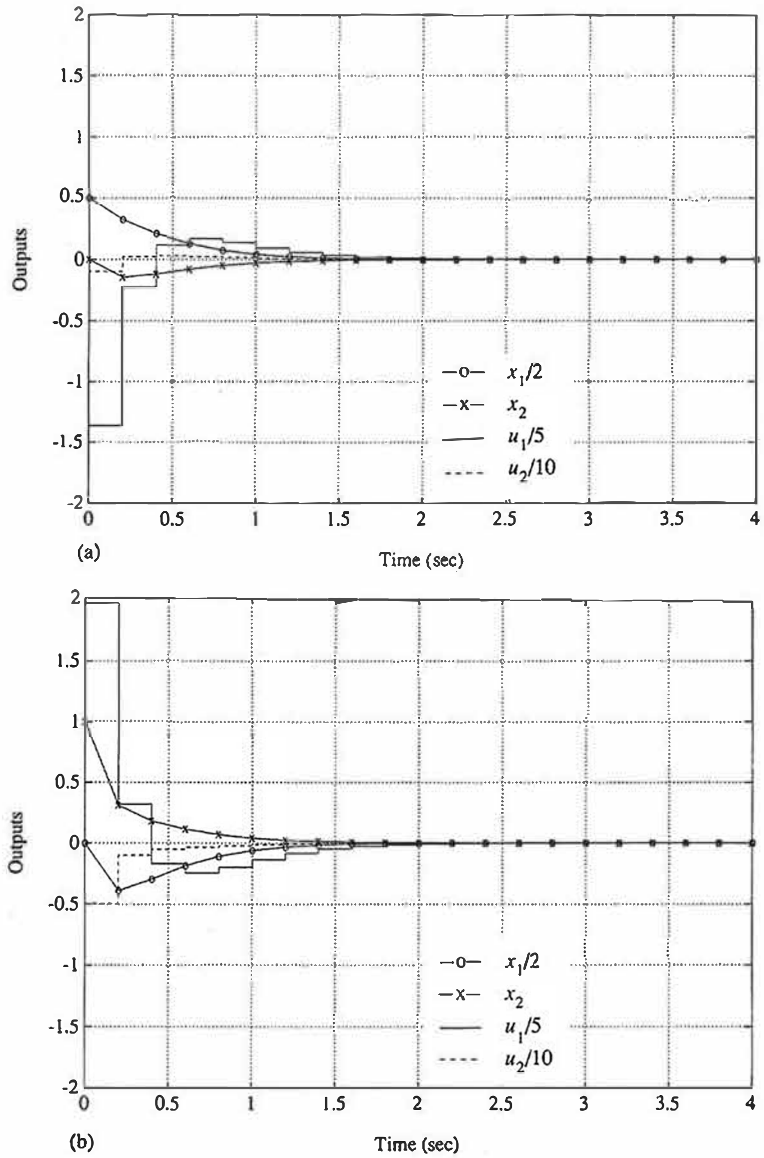


Figure 9.9 Response of paper-machine closed-loop control to: (a) initial state $x^T(0) = [1 \quad 0 \quad 0]$; (b) initial state $x^T(0) = [0 \quad 1 \quad 0]$

results in a design that still meets all specifications and substantially reduces the use of both controls. Its response to $\mathbf{x}(0) = [010]^T$ is shown in Figure 9.11.

Figure 9.10
Response of paper-machine closed-loop control to initial state $x^T(0) = [0 \quad 1 \quad 0]$ with t_s lengthened to 5 sec

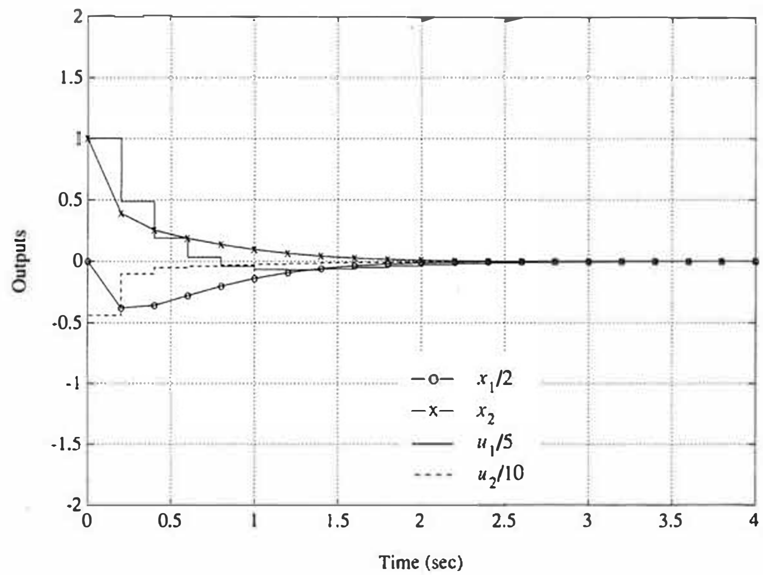
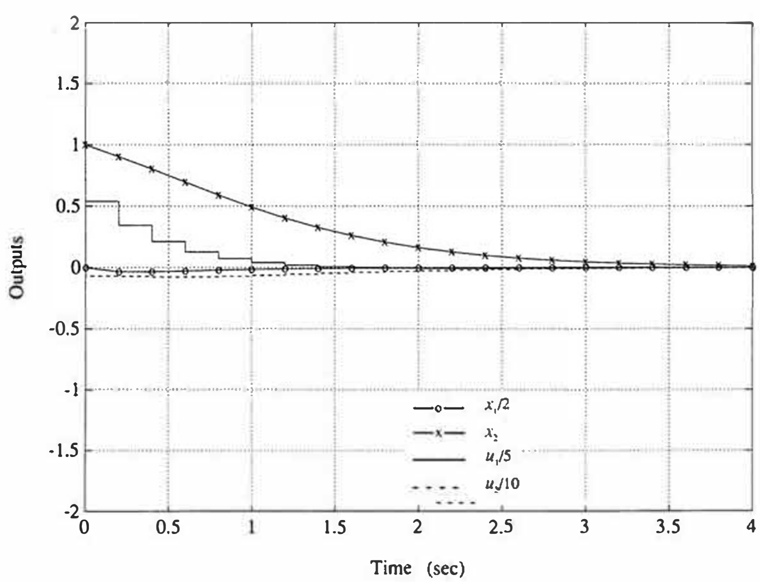


Figure 9.11
Response of paper-machine closed-loop control to initial state $x^T(0) = [0 \quad 1 \quad 0]$ with t_s lengthened to 5 sec and no weight on x_2



9.5.4 Magnetic-Tape-Drive Design Example

As a further illustration of MIMO design, we will now apply the ideas to an example that includes integral control and estimation. The system is shown in Fig. 9.12. There is an independently controllable drive motor on each end of the tape; therefore, it is possible to control the tape position over the read head, x_3, as well as the tension in the tape. We have modeled the tape to be a linear spring with a small amount of viscous damping. Although the figure shows the tape head to the left of the spring, in fact, the springiness is distributed along the full length of the tape as shown by the equations below. The goal of the control system is to enable commanding the tape to specific positions over the read head while maintaining a specified tension in the tape at all times. We will carry out the design using MIMO techniques applied to the full system equations and conclude the discussion by illustrating how one could perform the design using a decoupled model and SISO techniques.

The specifications are that we wish to provide a step change in position of the tape head, x_3, of 1 mm with a 1% settling time of 250 msec with overshoot less than 20%. Initial and final velocity of x_3 is to be zero. The tape tension, T_e, should be controlled to 2 N with the constraint that $0 < T_e < 4$ N. The current is limited to 1 A at each drive motor.

The equations of motion of the system are

$$\begin{aligned} J\ddot{\theta}_1 &= -T_e r + K_m i_1, \\ J\ddot{\theta}_2 &= -T_e r + K_m i_2, \\ T_e &= k\,(x_2 - x_1) + b\,(\dot{x}_2 - \dot{x}_1), \\ x_3 &= (x_1 + x_2)/2, \end{aligned} \tag{9.116}$$

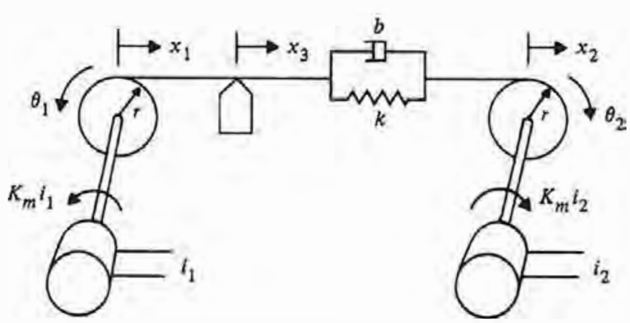


Figure 9.12
Magnetic-tape-drive design example

where

$$
\begin{aligned}
i_1, i_2 &= \text{current into drive motors 1 and 2, respectively (A)},\\
T_e &= \text{tension in tape (N)},\\
\theta_1, \theta_2 &= \text{angular position of motor/capstan assembly (radians)},\\
x_1, x_2 &= \text{position of tape at capstan (mm)},\\
x_3 &= \text{position of tape over read head (mm)},\\
J &= 0.006375\,\text{kg} - \text{m}^2,\ \text{motor and capstan inertia},\\
r &= 0.1\,\text{m},\ \text{capstan radius},\\
K_m &= 0.544\,\text{N} - \text{m/A},\ \text{motor torque constant},\\
k &= 2113\,\text{N/m},\ \text{tape spring constant, and}\\
b &= 3.75\,\text{N} - \text{sec/m},\ \text{tape damping constant.}
\end{aligned}
$$

In order to be able to simulate a control system design on inexpensive equipment where the system has time constants much faster than 1 sec, it is often useful to time-scale the equations so that the simulation runs slower in the laboratory than it would on the actual system.[17] We have chosen to time-scale the equations above so they run a factor of 10 slower than the actual system. Therefore, the settling time specifications become 2.5 sec instead of 250 msec for the actual system. The numerical equations below and all the following discussion pertain to the time-scaled system. Incorporating the parameter values and writing the time-scaled equations in state form results in

$$
\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{\omega}_1 \\ \dot{\omega}_2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -10 & 0 \\ 0 & 0 & 0 & 10 \\ 3.315 & -3.315 & -0.5882 & -0.5882 \\ 3.315 & -3.315 & -0.5882 & -0.5882 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \omega_1 \\ \omega_2 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 8.533 & 0 \\ 0 & 8.533 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \end{bmatrix}, \tag{9.117}
$$

where the desired outputs are

$$
\begin{bmatrix} x_3 \\ T_e \end{bmatrix} = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ -2.113 & 2.113 & 0.375 & 0.375 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \omega_1 \\ \omega_2 \end{bmatrix}, \tag{9.118}
$$

where

$$\omega_1, \omega_2 = \text{angular rates of motor/capstan assembly}, \dot{\theta}_i\ \text{(rad/sec)}.$$

The eigenvalues of the system matrix in Eq. (9.117) are

$$s \cong 0,\ 0,\ \pm j8\ \text{rad/sec},$$

17 See Franklin, Powell, and Emami-Naeini, 8th Ed (2019) for a discussion of time-scaling.

where the oscillatory roots are from the spring–mass system consisting of the tape and the motor/capstan inertias. This open-loop resonance has a higher frequency than the required closed-loop roots to meet the settling time specifications; therefore, it would be wise to sample at $15\times$ the open-loop resonance at 8 rad/sec. A sample rate of $T = 0.05$ sec results.

Full State Feedback

As a first step in the design, let's try state feedback of all four states in Eq. (9.117). We will address the state estimation in a few pages.

The output quantities to be controlled are x_3 and T_e so that it is logical to weight those quantities in the cost $\overline{\mathbf{Q}}_1$. According to the guidelines in Eq. (9.106), we should pick

$$\overline{\mathbf{Q}}_1 = \begin{bmatrix} 1 & 0 \\ 0 & \frac{1}{4} \end{bmatrix},$$

and because we expect each control to be used equally

$$\mathbf{Q}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Although these weightings gave an acceptable result, a somewhat more desirable result was found by modifying these weightings to

$$\overline{\mathbf{Q}}_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

where a third output has been added to that of Eq. (9.118), which represents $\dot{T}_e$ in order to obtain better damping of the tension. Thus we obtain the weighting matrix from

$$\mathbf{Q}_1 = \mathbf{H}_w^T \overline{\mathbf{Q}}_1 \mathbf{H}_w, \tag{9.119}$$

where

$$\mathbf{H}_w = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ -2.113 & 2.113 & 0.375 & 0.375 \\ 0 & 0 & 0.5 & 0.5 \end{bmatrix}. \tag{9.120}$$

Use of the $\mathbf{Q}_1$ and $\mathbf{Q}_2$ with the discrete equivalent of Eq. (9.117) in MATLAB's dlqr.m results in the feedback gain matrix

$$\mathbf{K} = \begin{bmatrix} -0.823 & 0.286 & 1.441 & 0.311 \\ -0.286 & 0.823 & 0.311 & 1.441 \end{bmatrix},$$

which can be used to form the closed-loop system for evaluation. Calculation of the eigenvalues of the discrete system and transformation of them to their discrete equivalent by $z = e^{sT}$ results in closed-loop roots at

$$s = -5.5 \pm j5.5, \; -10.4 \pm j12.4 \text{ sec}^{-1}.$$

To bring the system to the desired values of T_e and x_3, we use the state command structure described in Section 8.4.2. Let us first try without the feedforward value of steady-state control, u_{ss}. Calculation of the reference state, $\mathbf{x}_r$, can be carried out from the physical relationships that must exist at the desired values of $x_3 = 1$ mm and $T_e = 2$ N

$$x_1 + x_2 = 2x_3 = 2,$$

$$x_2 - x_1 = \frac{T_e}{k} = \frac{2}{2113} = 0.000947 \text{ m} = 0.947 \text{ mm}.$$

Solving these two equations and adding the zero velocities results in

$$\mathbf{x}_r = \begin{bmatrix} 0.527 \\ 1.473 \\ 0 \\ 0 \end{bmatrix}.$$

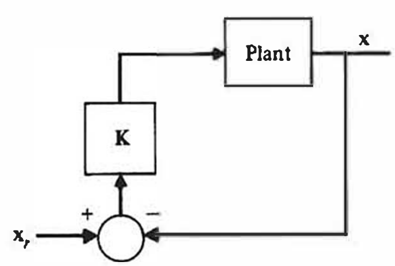
Evaluation of the time response using the structure in Fig. 9.13 (from Fig. 8.14) shows that the input current limits are violated and results in a settling time of 1 sec, which is significantly faster than required. In order to slow down the response and thus reduce the control usage, we should lower $\overline{\mathbf{Q}}_1$. Revising the weightings to

$$\overline{\mathbf{Q}}_1 = \begin{bmatrix} 0.1 & 0 & 0 \\ 0 & 0.1 & 0 \\ 0 & 0 & 0.1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \tag{9.121}$$

results in

$$\mathbf{K} = \begin{bmatrix} -0.210 & 0.018 & 0.744 & 0.074 \\ -0.018 & 0.210 & 0.074 & 0.744 \end{bmatrix},$$

Figure 9.13
Reference input structure used for Fig. 9.14



which produces a closed-loop system whose roots transform to

$$s = -3.1 \pm j3.1, -4.5 \pm j9.2 \text{ sec}^{-1},$$

and produces the response shown in Fig. 9.14. The figure shows that x_3 has a settling time of about 2 sec with an overshoot less than 20% while the control currents are within their limits; however, the tension did not go to the desired 2 N. The steady-state tension error of the system is substantial, and so we shall proceed to correct it.

As discussed in Section 8.4.2, we can provide a feedforward of the required steady-state value of the control input that eliminates steady-state errors. Evaluation of Eq. (8.73) (see refi.m in the Digital Control Toolbox) using $\mathbf{H}$ from Eq. (9.118) and $\mathbf{\Phi}$, $\mathbf{\Gamma}$ from Eq. (9.117) leads to

$$\mathbf{N}_x = \begin{bmatrix} 1 & -0.2366 \\ 1 & +0.2366 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{N}_u = \begin{bmatrix} 0 & 0.1839 \\ 0 & 0.1839 \end{bmatrix},$$

which can be used to command the system as in Fig. 8.15(a) with $\mathbf{r} = [\,1\,\text{mm} \quad 2\,\text{N}\,]^T$. Note that $\mathbf{N}_x\mathbf{r} = \mathbf{x}_r = [\,0.527 \quad 1.473 \quad 0 \quad 0\,]^T$ as already computed above, and we now have the steady state control, $\mathbf{u}_{ss} = \mathbf{N}_u\mathbf{r} = [\,0.368 \quad 0.368\,]^T$, that should eliminate the tension error. The response of the system with this modification is shown in Fig. 9.15. It verifies the elimination of the tension error and that all other specifications are met.

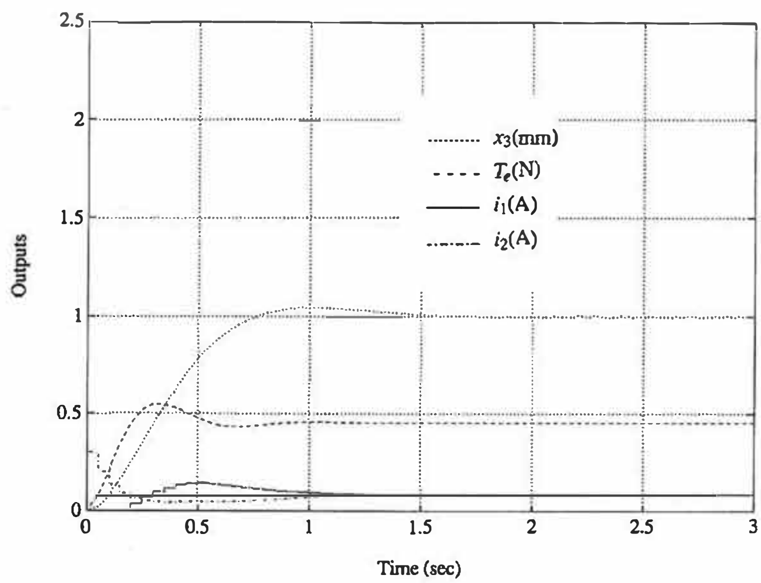


Figure 9.14
Time response using Eq. (9.121)

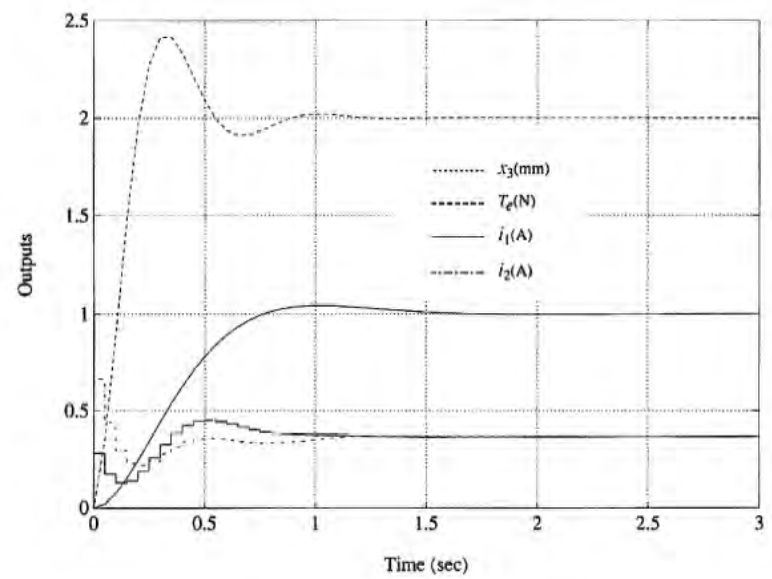


Figure 9.15
Time response using Eq. (9.121) and feedforward of u_{ss}

Normally, designers prefer to avoid the use of feedforward of $\mathbf{u}_{ss}$ alone because its value depends on accurate knowledge of the plant gain. The preferred method of reducing steady-state errors is to use integral control or possibly to combine integral control with feedforward. In the latter case, the integral's function is simply to provide a correction for a possible error in the feedforward and thus can be designed to have less of an impact on the system bandwidth. So let's design an integral control to replace the $\mathbf{u}_{ss}$ feedforward.

Because the tension was the only quantity with the error, the integral control need only correct that. Therefore, we will integrate a measurement of the tension and use it to augment the feedback. We start out by augmenting the state with the integral of T_e according to Eq. (8.83) so that

$$\begin{bmatrix} x_I \\ \mathbf{x} \end{bmatrix}_{k+1} = \begin{bmatrix} 1 & \mathbf{H}_I \\ \mathbf{0} & \mathbf{\Phi} \end{bmatrix} \begin{bmatrix} x_I \\ \mathbf{x} \end{bmatrix}_k + \begin{bmatrix} 0 \\ \mathbf{\Gamma} \end{bmatrix} u(k), \tag{9.122}$$

where, from Eq. (9.118)

$$\mathbf{H}_I = [\,-2.113 \quad 2.113 \quad 0.375 \quad 0.375\,].$$

We will use the augmented system matrices ($\mathbf{\Phi}_a$, $\mathbf{\Gamma}_a$) from Eq. (9.122) in an LQR computation with a revised $\mathbf{Q}_1$ that also weights the integral state variable. This is most easily accomplished by forming a revised $\mathbf{H}_w$ from Eq. (9.120) that includes a fourth output consisting of the tension integral, x_I, and that is used with the augmented state defined by Eq. (9.122). The result is

$$\mathbf{H}_w = \begin{bmatrix} 0 & 0.5 & 0.5 & 0 & 0 \\ 0 & -2.113 & 2.113 & 0.375 & 0.375 \\ 0 & 0 & 0 & 0.5 & 0.5 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}. \tag{9.123}$$

The revised weighting matrices are then obtained by the use of Eq. (9.119) with an additional diagonal term in $\overline{\mathbf{Q}}_1$ corresponding to an equal weight on x_I, that is

$$\overline{\mathbf{Q}}_1 = \begin{bmatrix} 0.1 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 \\ 0 & 0 & 0.1 & 0 \\ 0 & 0 & 0 & 0.1 \end{bmatrix} \quad \text{and} \quad \mathbf{Q}_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{9.124}$$

Performing the calculations results in an augmented feedback gain matrix

$$\mathbf{K}_a = \begin{bmatrix} 0.137 & -0.926 & 0.734 & 1.26 & 0.585 \\ 0.137 & -0.734 & 0.926 & 0.585 & 1.26 \end{bmatrix}, \tag{9.125}$$

which can be partitioned into two gains, one for the integral and one for the original state

$$\mathbf{K}_I = \begin{bmatrix} 0.137 \\ 0.137 \end{bmatrix} \quad \text{and} \quad \mathbf{K} = \begin{bmatrix} -0.926 & 0.734 & 1.26 & 0.585 \\ -0.734 & 0.926 & 0.585 & 1.26 \end{bmatrix}. \tag{9.126}$$

These feedback gain matrices are then used to control the system as shown in Fig. 9.16, which is based on the ideas from Fig. 8.23. Essentially, the only difference between this implementation and the previous one is that we have replaced the $\mathbf{u}_{ss}$ feedforward with an integral of the tension error.

The roots of the closed-loop system transformed to the s-plane are

$$s = -9.3, -3.1 \pm j3.1, -5.8 \pm j11.8 \text{ sec}^{-1},$$

and the time response of the system is shown in Fig. 9.17. Again, the tension goes to the desired 2 N, as it should, and all other specifications are met.

Add the Estimator

The only unfinished business now is to replace the full state feedback with an estimated state. We will assume there are two measurements, the tension, T_e, and tape position, x_3. Although measurements of θ_1 and θ_2 would possibly be easier to make, it is typically best to measure directly the quantities that one is interested in controlling. Therefore, for purposes of estimation, our measurement matrix is

$$\mathbf{H}_e = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ -2.113 & 2.113 & 0.375 & 0.375 \end{bmatrix}. \tag{9.127}$$

Figure 9.16
Integral control implementation with reference input and full state feedback

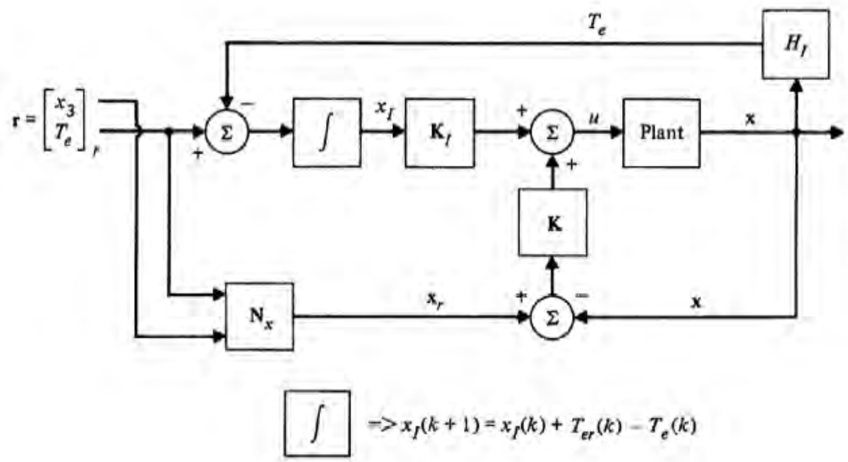
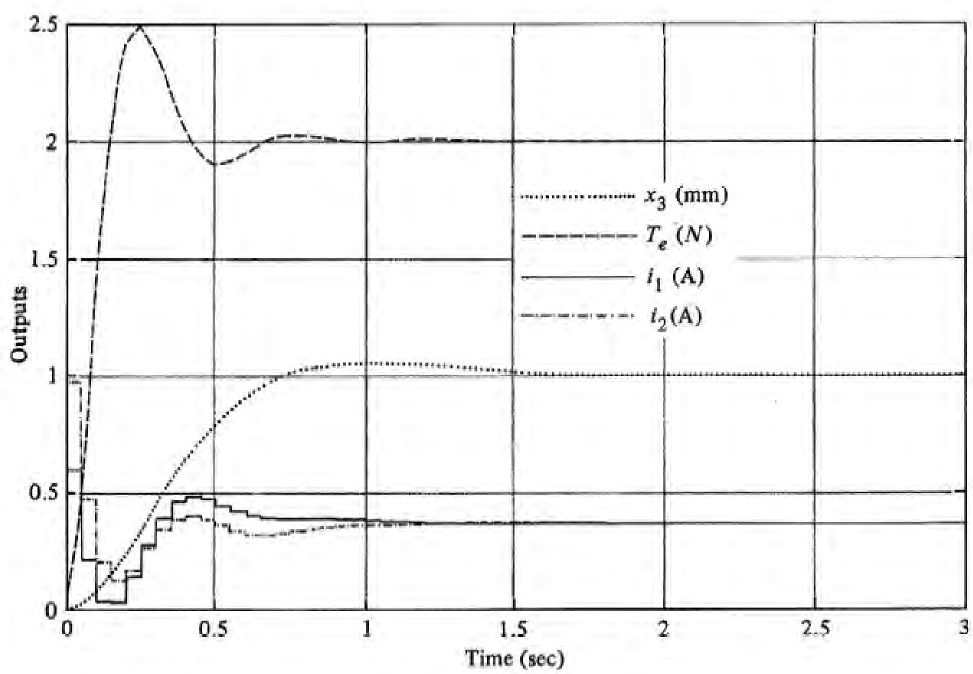


Figure 9.17
Time response using Eq. (9.126) and integral control as in Fig. 9.16



Let us suppose that the rms accuracy of the tape position measurement is 0.02 mm. According to Eq. (9.104), the first diagonal element of the $\mathbf{R}_v$ matrix is therefore $0.02^2 = 0.0004$. Let us also suppose that the rms accuracy of the tension measurement is 0.01 N. This results in

$$\mathbf{R}_v = \begin{bmatrix} 0.0004 & 0 \\ 0 & 0.0001 \end{bmatrix}. \tag{9.128}$$

Determining values for the process-noise matrix, $\mathbf{R}_w$, is typically not based on the same degree of certainty as that possible for $\mathbf{R}_v$. We will assume that the process noise enters the system identically to i_1 and i_2; therefore, $\mathbf{\Gamma}_1 = \mathbf{\Gamma}$ in Eq. (9.81). We could make measurements of the magnitude and spectrum of the input-current noise to determine $\mathbf{R}_w$ via Eqs. (9.101) and (9.102). However, other disturbances and modeling errors typically also affect the system and would need to be quantified in order to arrive at a truly optimal estimator. Instead, we will somewhat arbitrarily pick

$$\mathbf{R}_w = \begin{bmatrix} 0.0001 & 0 \\ 0 & 0.0001 \end{bmatrix}, \tag{9.129}$$

compute the estimator gains and resulting roots, then modify $\mathbf{R}_w$ based on the estimator performance in simulations including measurement noise and plausible disturbances. Proceeding, we use the unaugmented $\mathbf{\Phi}$ and $\mathbf{\Gamma}_1$ based on Eq. (9.117) in MATLAB's kalman.m with $\mathbf{H}_e$ from Eq. (9.127) and the noise matrices from Eqs. (9.128) and (9.129) to find that

$$\mathbf{L} = \begin{bmatrix} 0.321 & -0.120 \\ 0.321 & 0.120 \\ -0.124 & 0.142 \\ 0.124 & 0.142 \end{bmatrix}, \tag{9.130}$$

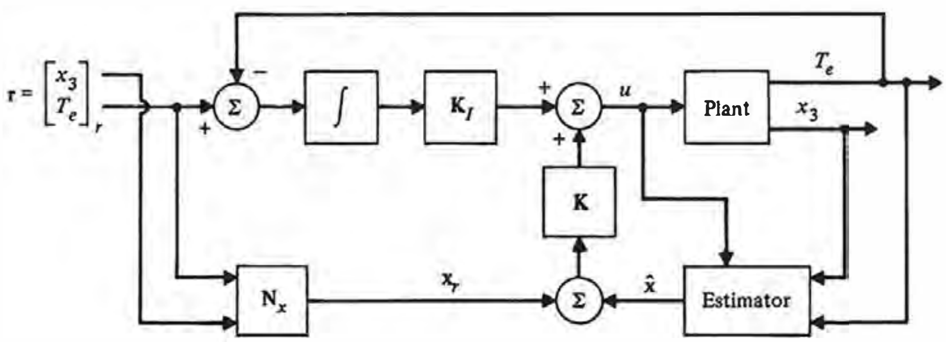
which results in estimator error roots that transform to

$$s = -3.0 \pm j4.5,\ -4.1 \pm j15.4 \text{ sec}^{-1}. \tag{9.131}$$

If implemented using the state command structure as in Fig. 9.18, which has the same reference input structure as in Fig. 9.16, the response will be identical to Fig. 9.17. This follows because the estimator model receives the same control input that the plant does and thus no estimator error is excited. In order to evaluate the estimator, we need to add measurement noise and/or some plant disturbance that is not seen by the estimator. Figure 9.19 shows the system response to the same reference input as Fig. 9.17; but zero mean random noise with an rms of 0.02 mm has been added to the x_3 measurement, noise with an rms of 0.01 N has been added to the T_e measurement, and a step disturbance of 0.01 A has been added to the i_1 entering the plant[18] at 1.5 sec. Note in the figure that there is

18 Only the plant, not the estimator.

Figure 9.18
Integral control implementation with reference input and estimated state feedback



$$\boxed{\int} \Rightarrow x_I(k+1) = x_I(k) + T_{er}(k) - T_e(k)$$

Figure 9.19
Time response with integral control and an estimator with L from Eq. (9.130) including measurement noise and a disturbance at 1.5 sec